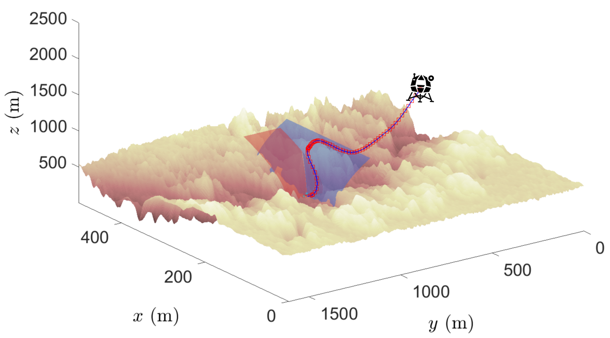
DEPARTMENT OF AEROSPACE ENGINEERING
INDIAN INSTITUTE OF TECHNOLOGY MADRAS
CHENNAI – 600036

Near-Fuel-Optimal Guidance for Terrain-Avoided Planetary Landing



A Thesis

Submitted by

SHEIKH ZEESHAN BASAR

For partial fulfilment of the degree

Of

MASTER OF SCIENCE BY RESEARCH

May 2024

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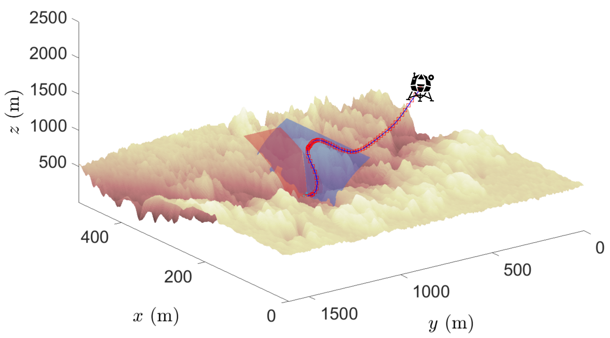

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Work well, study well,

Play well, eat well, and sleep well!

This is the way of the Kamesen style!

– Master Roshi

To my precious family.

THESIS CERTIFICATE

This is to undertake that the Thesis titled **NEAR-FUEL-OPTIMAL GUIDANCE FOR TERRAIN-AVOIDED PLANETARY LANDING**, submitted by me to the Indian Institute of Technology Madras, for the award of **Master of Science by Research**, is a bona fide record of the research work done by me under the supervision of **Dr. Satadal Ghosh**. The contents of this Thesis, in full or in parts, have not been submitted to any other Institute or University for the award of any degree or diploma.

Chennai 600036 **Sheikh Zeeshan Basar**

Date: 31 May 2024

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IIT Madras

LIST OF PUBLICATIONS

I. PUBLICATIONS IN INTERNATIONAL CONFERENCE PROCEEDINGS

Basar, Sheikh Zeeshan, and Satadal Ghosh. "Fuel-Optimal Powered Descent Guidance for Hazardous Terrain." *IFAC-PapersOnLine*, vol. 56, no. 2, 2023, pp. 6018–23.

II. PUBLICATIONS UNDER REVIEW IN INTERNATIONAL JOURNAL

Basar, Sheikh Zeeshan, and Satadal Ghosh. "Robust Fuel-Optimal Landing Guidance for Hazardous Terrain Using Multiple Sliding Surfaces." *arXiv:2403.12584 [eess.SY]*, 2024.

III. MANUSCRIPT UNDER PREPARATION FOR INTERNATIONAL JOURNAL

Basar, Sheikh Zeeshan, and Satadal Ghosh. "Robust and Near-Fuel-Optimal Landing Guidance with Online Terrain Barrier Update Feature".

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People often say you are a mosaic of everyone you have ever spent time with. The existence of this thesis is proof of that.

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People who know me from before my time at IIT Madras will say that I don't easily make friends. But with some twist of fate, I found a group of talented, brilliant and, most importantly, kind people who have become nothing short of a chosen family to me. To Shahid, Arunima, Akash, Deepthy, and Fahim: Thank you for being part of my survival kit. There is no way I could have done this in Heaven and Earth if you weren't there. I will forever cherish the joys and the sadness we faced together. I will miss the gym, the track, the pool, almost an infinite amount of BBQ Spicy Chicken (shout out to Zaitoon) and the tea and *tea* sessions. A big thank you to Bhumika, Naman, Jannie, Koushik, Navaneetha, Sneha, Gayathri, Kartik, Avik, Prabhakar, Wishu and everyone else in the extended “Madras Gang”. Sorry if I missed your name. The outings, the sweets, and the birthdays were all incredible because of all of you.

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ABSTRACT

KEYWORDS Spacecraft guidance, Autonomous planetary landing, Terrain avoidance, Fuel optimal guidance, Robustness

Interplanetary missions carry expensive and sensitive equipment for conducting experiments. To ensure that a planetary exploration mission is a success, all the payloads must survive the landing, that is, the landing must be precise and soft. Further, the ability of the spacecraft to manoeuvre around hazardous terrain on its way to the desired landing site becomes a vital mission aspect. However, large diverts and manoeuvres consume a significant amount of fuel, leading to less fuel for emergencies or return missions, therefore requiring more fuel to be carried onboard. Further, it is of utmost importance that the spacecraft motion be robust to external disturbance (e.g. winds) and uncertainties (e.g. thrust lag and actuation noise) and be able to land softly at the desired landing site with a high degree of accuracy. Finally, the guidance law should be such that the spacecraft should rely on onboard sensors to determine nearby hazardous terrain and avoid them on its way to the desired landing site.

This thesis presents optimal control-based guidance laws for terrain-avoided precision soft landing of spacecraft in a nonlinear setting to achieve the motivations above. First, the Optimal Terrain Avoided Landing Guidance (OTALG) law is derived by solving the fuel optimality problem by minimising the performance index augmented with a penalty function of the spacecraft's distance to the terrain barriers. These terrain barriers are polynomial functions determined beforehand using a step-shaped model of the terrain from the orbital images. Using extensive simulations and statistical studies, the OTALG law was shown to be near-fuel-optimal compared to the current state-of-the-art guidance law for terrain-avoided, fuel-optimal soft landing.

Second, the OTALG law is made robust against external disturbances and uncertainties by using sliding mode control with multiple sliding surfaces (MSS). The presented guidance

law, named 'MSS-OTALG', improves the landing performance in terms of precision soft landing accuracy. Further, the sliding parameter is designed to allow the lander to avoid terrain by leaving the trajectory enforced by the sliding mode, and eventually returning to it when the terrain avoidance phase is completed. And finally, the robustness of the MSS-OTALG is established by proving practical fixed-time stability. Extensive numerical simulations are also presented to showcase its performance in terms of terrain avoidance, low fuel consumption and accuracy of precision soft landing. Comparative studies against relevant literature validate a balanced trade-off of all these performance measures achieved by the developed MSS-OTALG.

Third, this thesis proposes an algorithm titled "Online Terrain Barrier Updates" (OTBU) to further improve the autonomy of the landing guidance by relying on the onboard sensor measurements to generate the terrain barriers instead of relying on the pre-computed terrain barriers. The proposed algorithm uses pre-processed elevation maps, state measurements from the onboard sensors, and an estimate of the highest peak in the neighbourhood of the landing site from orbital images to generate terrain barriers. OTBU uses simple peak detection methods from classical image processing to analyse the elevation data and determine the shape of the barriers. This, along with adaptive barrier update interval, results in low computational load while ensuring the spacecraft is safely steered away from the hazardous terrain. The barriers are then integrated with MSS-OTALG for a precision soft landing. The terrain avoidance and landing performance are validated using extensive simulations using real Martian elevation maps from NASA's HiRISE database. Soft landing precision and fuel consumption are validated using Monte Carlo-like simulations, and quantitative results are presented.

CONTENTS

LIST OF TABLES

LIST OF FIGURES

CHAPTER 1

INTRODUCTION

Under the backdrop of the Cold War, from the late 1940s to the late 1980s, the USA and the erstwhile USSR raced to not only develop nuclear weapons and attempt to gain a lasting advantage in the global theatre, but both superpowers raced to be the first in the space as well. Space Race, as it was called, seeded the development of many technologies we take for granted today, such as GPS, global communication, remote sensing, etc. Among all these, scientists worldwide were excited about reaching the celestial bodies that dot the heavens. The achievement of being the first man-made object to reach a celestial body belongs to Luna 2, [35] a part of the Luna series of missions spearheaded by the erstwhile USSR. In addition to transmitting data from several payloads such as ion traps, radiation detectors, etc., the success of this spacecraft was considered a massive cultural win for the Soviet Union. The USA was, however, not far behind. A few years and several failures later, Ranger 7 [35] was the first successful American lunar mission. The primary objective of the Ranger programs was to take closeup pictures of the lunar surface and transmit them back until the spacecraft crashed on the lunar surface and was destroyed. However, most spacecraft landing missions intend to function and observe various parameters of interest over an extended period, and for that, they must first survive the landing. Thus, building upon the technological demonstrations of Luna and Ranger missions, studies to achieve soft landing on celestial bodies began with a renewed interest.

The key objective of planetary landing missions is to collect and analyse samples and relay the data to Earth. For this, the lander needs to carry an extensive set of scientific modules, avoid any damage during the entry, descent and landing (EDL) phase and land as close to the area of interest as possible, even under disturbances and uncertainties, to get the most out of these missions. Interplanetary missions in the near future will carry more and more valuable equipment and may need to land near hillocks or cliff faces. This necessarily requires a landing guidance algorithm that is not only accurate in precision soft landing, robust to disturbances, simple to implement, and fuel optimal but can also successfully perform terrain avoidance as well.

To this end, this thesis will attempt to bridge the gap in the current state-of-the-art and develop a simple-to-implement and autonomous guidance framework for safe and soft landing of spacecraft in hazardous terrain, which is close to fuel optimal, robust against disturbances and is highly accurate.

1.1 LITERATURE REVIEW

The study of autonomous guidance laws for powered descent began with the Apollo era. However, most guidance laws were rudimentary, as they were severely limited by the computational power and memory. In fact, the Apollo missions used a straightforward polynomial-based guidance law [25]. The polynomial-based guidance is relatively easier to implement; however, back in the 1950s they lacked in fuel-optimality, robustness and the required accuracy. However, using the technology developed in the last 50 years or so, modern Mars missions such as Curiosity, Tianwen-1, and Perseverance have successfully been able to reduce the landing error to less than 10km [29; 40].

1.1.1 Numerical Methods

Since the late 1990s, significant efforts have been made to improve fuel efficiency. As computing power readily increases, numerical optimisation methods have come into the spotlight. Convex optimisation methods have gained popularity in both research and industry due to the theoretical guarantees of optimality and convergence, and the computational performance. However, realistic missions have non-convexities due to thrust constraints, nonlinear dynamics, and coupled state-control dynamics. To address these, algorithms such as Lossless Convexification (LCvx) and Successive Convexification

(SCvx) were been developed [1; 5; 6; 27]. In LCvx, the general non-convex problem is first convexified using a slack variable, and then the globally optimal solution of the relaxed problem is recovered using Pontryagin's maximum principle. However, LCvx can only be applied under very specific cases. When LCvx cannot be applied, convex optimisation can still be applied by using sequential convex programming (SCP) in SCvx. In SCvx, all the non-convexities are linearised and the resulting convex problem is solved in the local neighbourhood. This is an iterative, trust region-based method, where the whole process is repeated till desired convergence is reached. SCvx [27] has been used successfully for the powered decent problem [37]. Both these algorithms have been successfully used for powered descent problems in high-fidelity simulations and in real-life experiments [26].

1.1.2 Feedback Guidance

Convex optimisation-based methods are open-loop and, therefore, highly susceptible to perturbations and estimation errors. The above mentioned issues can be alleviated by using the feedback guidance laws presented in the relevant literature. Classical feedback laws for missile guidance, like Proportional Navigation Guindance (PNG) as discussed in [42], have inspired some of the earliest guidance laws for autonomous landing. Several variations of PNG exists and its adaptations such as biased-PNG [24] and pulsed-PNG [39], have been explored for powered descent. The concept of zero-effort-miss (ZEM) is extensively used in missile guidance, which denotes the miss distance from the desired terminal position if no control effort is applied from the current time forward. The idea of ZEM was extended in [10] to include the deviation in terminal velocity, zero-effort-velocity (ZEV). Then using results from optimal control, Optimal Guidance Law (OGL) was developed as a function of ZEM and ZEV [10].

A critical drawback of the vanilla OGL is that a part of the trajectory maybe underground for a large terminal time implying a crash landing at an undesired location, thus defeating the purpose of the mission. While lowering terminal time may help in avoiding this issue, it requires a significantly larger thrust command to succeed. Reference [19] suggested use of waypoints to avoid crashing into the planetary surface, while selection of such waypoints optimally was studied by [20]. However, the method developed there becomes computationally expensive as the number of time-steps increases and if both terminal and waypoint times are left free for the optimiser to solve.

1.1.3 Sliding Mode Control

Utilising the core idea of ZEM/ZEV-based optimal guidance law, powered descent was made robust against perturbations using Sliding Mode Control, and its variants [12–14]. Further, missile guidance concepts, such as collision course and heading error, further inspired the work in [33] to develop a novel sliding mode control-based (SMC) guidance law in full nonlinear setting which was then extended to include fuel optimality as well in [34]. Due to its effectiveness, multiple sliding surfaces (MSS) have also been used to improve the robustness against external disturbances. MSS has attracted much attention for space applications in the recent past. For example, MSS has been used in [12] for precision landing in asteroids as well as for autonomous landing on Mars as described in [15]. In the presence of atmospheric disturbances, an optimal sliding guidance (OSG) presented in [38] was found to perform well with high degree of precision for soft landing even with partial loss of thrust. The guidance laws proposed in [12; 15; 38] have been proved to be finite time stable (FTS) as well.

1.1.4 Terrain Avoidance

Note that the guidance laws mentioned above either did not consider to avoid the terrain or used simple glideslopes or glideslope-like constraints to avoid crashing into the terrain. More dedicated studies on terrain avoided landing has also been presented in recent literature. For example, the results of LCVx were extended to incorporate terrain avoidance in fuel-optimal powered descent phase in [2]. But, the proposed algorithm therein is computationally heavy and has an open-loop structure, thus making it susceptible to disturbances and hence lacking in robustness. In order to avoid collision during the powered descent phase, an improved ZEM/ZEV feedback guidance was presented in [45] by including a switching-form term as penalty in the performance index, which required manual tuning to achieve desired performance. The limitation posed by dependence on prior tuning in the added term therein was subsequently obviated in [43] by using a self-adjusting augmentation to the performance index. Using Barrier Lyapunov functions in [16] and Prescribed Performance functions in [17], guidance laws for terrain-avoided soft landing were proposed. Both the guidance laws were able to manoeuvre to avoid the terrain and soft land at the desired landing point, however, both of them required several difficult-to-estimate time-dependent variables to generate the terrain bounding barriers. Additionally, these approaches did not consider the aspect of fuel efficiency and achieving satisfactory precision performance within thrust constraints while designing the guidance laws.

1.1.5 Vision-based Terrain Navigation and Guidance

One of the earliest work on terrain mapping for planetary mission was done in [21]. Here, for a roving vehicle, an algorithm to generate an elevation map using a single range image, and a method to determine transformation between two images using terrain features are proposed. Since then several advancements have been made in not only developing terrain mapping methods for ground robots, but also for aerial vehicles.

One of the most popular method for terrain navigation is to use a single downward-pointing camera along with efficient implementation of SLAM algorithms for accurate localising the vehicle, detect and avoid hazardous terrain, and perform autonomous and safe landing [7]. Given a set of two-dimensional images taken from a moving camera, the three-dimensional structures can be reconstructed using Structure from Motion (SfM) technique is used. Reference [23] used an efficient implementation of SfM along with altimeter measurements to generate dense elevation map from a single camera. To improve resource efficiency, [32] proposed an SfM method that utilised state estimates in a tightly coupled manner to generate dense, accurate, hierarchical multi-resolution depth map which is sufficient in many outdoor manoeuvres and landing site detection.

In the contemporary literature, the primary focus is on the navigation aspect and the guidance part is either done by simplified reference trajectory tracking [18] or by using waypoint tracking [8; 11]. For trajectories for reference tracking are generated at fixed, discrete epoch. Similarly in the waypoints tracking techniques, the waypoints are generated as discrete events. Further, the aforementioned techniques relying on SfM are very resource-heavy as well. In the literature, there are very few papers which discuss the problem of integrated navigation and guidance for terrain-avoided precision soft landing. The factor that distinguishes the integrated navigation and guidance method over the navigation and reference/waypoint tracking method is the ability of the former to continuously update terrain information and accordingly guide the spacecraft to the desired landing site in a hazard avoided way. In [41], a hazard avoidance guidance law for landing on asteroids was presented which utilise super-twisting sliding mode control. Here, a repulsive potential function is used to avoid the hazardous terrain, however, designing the potential function requires prior knowledge of hazardous terrains' location. To reduce the computational burden associated with obtaining high resolution elevation maps around the desired landing site and its neighbouring terrain, [44] proposes a method to use dynamic low resolution RADAR images during descent stage for obstacle avoidance. While this effectively reduces the computational burden, the landing site selection is limited to flat, crater-like landing site, effectively reducing the possible locations where the spacecraft can land.

1.2 MOTIVATION AND OBJECTIVES

The general spacecraft landing problem is well discussed problem in the available literature. Several state-of-the-art guidance techniques have been shown to have excellent performance and have been flight proven. However, the *terrain-avoided* soft landing problem has not been discussed in much depth. The papers which address this issue have done so by using glideslope, or glideslope-like constraints. More dedicated studies have been performed, but they lack in the discussions about the fuel-optimality. To bridge this gap, a guidance formulation in nonlinear framework need to be developed to manoeuvre the spacecraft away from the terrain and softly land near the desired landing site while having low fuel consumption.

- To that end, the objective of Chapter 2 is to first, formulate safe zones for the spacecraft trajectories based on predetermined terrain information in the neighbourhood of the desired landing site, Then, develop a near-fuel-optimal guidance law by solving optimal control problem of minimising the control effort, which is penalised if the trajectories move close to the boundary of the safe zones. Further, validate the effectiveness developed guidance laws using extensive simulations and statistical methods.

Next, along with having high accuracy of precision soft landing, to ensure that the spacecraft survives the landing even under disturbances (such as those due to winds) and non-ideal guidance commands (such as those caused by thrust lags and actuation noise), the developed guidance law must be made robust and stable against these disturbances.

- The objective of Chapter 3 is to then expand the developed guidance law to incorporate globally stabilising control technique, such as SMC, to effectively reject disturbances and improve the accuracy of the precision soft landing which are validated using extensive simulations and statistical methods.

Relying solely on the predetermined terrain information in the neighbourhood of the desired landing site reduces autonomy of the spacecraft. Further, prior to the mission, the terrain information may not be reliably available to generate the safe zone. Therefore, the spacecraft must be able to use its onboard sensing and computation capability to generate the guidance commands to manoeuvre away from the hazardous terrain on its way to the desired landing site.

- To achieve this, the objective of Chapter 4 is to develop a method to generate the safe zones for the spacecraft motion on the fly using the onboard sensors instead of relying on predefined safe zones. Further, the method should be simple-to-implement and be able to perform well with low computational resources. And finally, test the effectiveness of the proposed algorithm, using extensive simulations and statistical methods, by integrating the proposed method with the guidance law previously developed.

1.3 SCOPE OF THESIS

To achieve the objective described in the previous section, the contributions of this thesis is as follows:

1. The predetermined terrain is first bounded using barrier polynomials which are parameterised by dimensions of a step-shaped model of the terrain. A novel augmentation to the standard minimum-fuel performance index is introduced next, which introduces penalty if the spacecraft moves close to the barriers. Using the augmented performance index and the terrain barriers, Optimal Terrain Avoidance Landing Guidance (OTALG) is developed as a function of ZEM and ZEV. Both of which can be calculated online, given the spacecraft's current position and velocity, time-to-go, and the terminal conditions.

2. Expanding upon OTALG, a robust guidance law, named MSS-OTALG, is developed by leveraging multiple sliding surface, where the first surface is used to track the positional error with a virtual controller introduced to ensure convergence of this sliding variable, while the second sliding surface is implemented to ensure that the first sliding variable follows the virtual control. To allow the spacecraft to navigate around rough terrain still maintain overall stability, the sliding parameter is suitably varied based on the system's states and time-to-go. Furthermore, with this selection of the sliding parameter, the practical fixed-time stability (PFTS) [22; 31] of the proposed MSS-OTALG is also established.

3. Next, to improve the autonomy of the proposed guidance laws an algorithm is proposed to generate the barrier functions on the fly. The proposed algorithm, titled "Online Terrain Barrier Update" (OTBU), ingests pre-processed terrain information, registered to the world coordinates, from a single downward-pointing camera and uses classical image processing techniques to detect peaks in the terrain to generate the terrain barriers. These are then fed into MSS-OTALG law to generate the guidance commands to drive to achieve safe and precision soft landing of the spacecraft.

4. Validate and verify all the proposed methods using extensive simulations and statistical methods.

1.3.1 Layout of the thesis

The thesis is organised in the following manner: Chapter 2 first introduces the concept of barrier functions to bound the predetermined terrain in the context of this thesis. After which a novel augmentation to the 2-norm performance index is presented, which is a function of the distance of the spacecraft to the barrier functions. Finally, a near-fuel optimal guidance law is obtained and the effectiveness of which is established using simulations and statistical methods. In Chapter 3, the drawbacks of the proposed near-optimal guidance law (such as lack robustness, poor accuracy and disturbance rejection) is alleviated by using SMC. The thus developed robust guidance law is again tested for its effectiveness using simulations and statistical methods. In Chapter 4, the requirement of knowing the terrain beforehand is removed and an algorithm to generate the barrier functions on the fly is presented. The proposed algorithm uses the onboard sensing equipment to estimate the terrain features and generate barrier functions as necessary. This chapter is then concluded with computer simulations on actual terrain maps. Finally, this thesis is concluded in Chapter 5 and some directions for future research are proposed.

CHAPTER 2

FUEL-OPTIMAL POWERED DESCENT GUIDANCE FOR HAZARDOUS TERRAIN

2.1 PROBLEM FORMULATION AND PRELIMINARIES

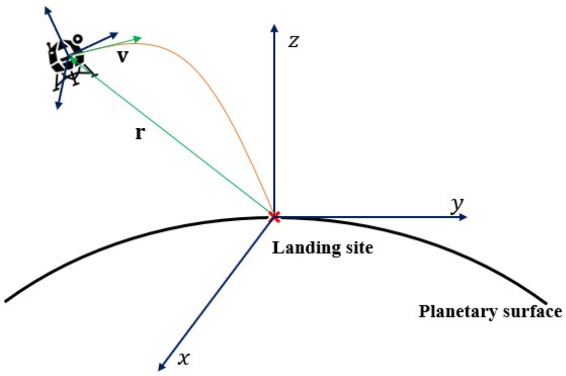


Figure 2.1: Spacecraft Landing Geometry

2.1.1 Problem formulations

The non-rotating inertial East-North-Up (ENU) frame with origin at the landing site is considered as shown in Fig. 2.1. Assuming a 3-DoF equations of motion, the lander can be modelled as:

$$\begin{aligned} \dot{\mathbf{r}} &= \mathbf{v} \\ \dot{\mathbf{v}} &= \mathbf{a} + \mathbf{g} + \mathbf{a}_{\mathrm{p}} \\ \mathbf{a} &= \frac{\mathbf{T}}{m} \\ \dot{m} &= -\frac{\|\mathbf{T}\|}{I_{\mathrm{sp}} g_e} \end{aligned} \tag{2.1}$$

where, $\mathbf{r} = [r_x, r_y, r_z]^{\mathrm{T}}$ and $\mathbf{v} = [v_x, v_y, v_z]^{\mathrm{T}}$ represent the position and velocity of spacecraft, and $\mathbf{g} = [g_x, g_y, g_z]^{\mathrm{T}}$ is the local gravity. The guidance command provided by the thrusters is represented by $\mathbf{a} = [a_x, a_y, a_z]^{\mathrm{T}}$, $\mathbf{a}_p$ is the net acceleration caused due to bounded perturbations (e.g. wind), m is the lander's mass, I_{sp} is the specific impulse, and g_e is the gravitational acceleration of Earth. Since the altitude at which the powered descent stage starts is significantly smaller than the radius of the planets, it is valid to assume a constant gravitational acceleration acting only along the z-axis, that is, $\mathbf{g} = [0, 0, -g]^{\mathrm{T}}$.

2.1.2 Preliminaries

The core idea of ZEM and ZEV is to determine the deviation from the desired position ($\mathbf{r}_f^d$) and velocity ($\mathbf{v}_f^d$) at the final time t_f if no control effort is applied from current time t. Defining time-to-go, $t_{\mathrm{go}} \triangleq t_f - t$, we have:

$$\begin{aligned} \mathbf{ZEM}(t) &= \mathbf{r}_f^d - \left[\mathbf{r}(t) + \mathbf{v}t_{\mathrm{go}} + \tfrac{1}{2}\mathbf{g}t_{\mathrm{go}}^2\right] \\ \mathbf{ZEV}(t) &= \mathbf{v}_f^d - \left[\mathbf{v}(t) + \mathbf{g}t_{\mathrm{go}}\right] \end{aligned} \tag{2.2}$$

where, $\mathbf{r}_f^d$, and $\mathbf{v}_f^d$ represent the desired position and velocity at the final time (t_f). The classical ZEM/ZEV-based optimal guidance law of [10] was derived by considering the performance index,

$$J = \frac{1}{2}\int_{t_0}^{t_f} \mathbf{a}^{\mathrm{T}}\mathbf{a}\, d\mathrm{t}, \tag{2.3}$$

subjected to the (2.1), with $\mathbf{a}_{\mathrm{p}} = 0$. The optimal guidance law was obtained there as,

$$\mathbf{a} = \frac{6}{t_{\mathrm{go}}^2}\mathbf{ZEM}(t) - \frac{2}{t_{\mathrm{go}}}\mathbf{ZEV}(t). \tag{2.4}$$

2.2 BARRIER DEFINITION AROUND HAZARDOUS TERRAIN

Terrain data available from reconnaissance missions are considered to be used to create the n number of step-shaped polygons to approximate the terrain, as shown in Fig. 2.2. Defining the height of j^{th} step as $h_{i,j}$, and the horizontal distance from the landing site (origin) along i-axis as $w_{i,j}$, where $i = x,\ y$.

2.2.1 Barriers for horizontal motion

To restrict the spacecraft motion in the horizontal planes, a barrier polynomial $\rho_{i,j}$ is defined as in (2.5). For n-steps, $n + 1$ number of barriers are defined, where the order of the polynomial is determined on the basis of degree of conservatism the mission

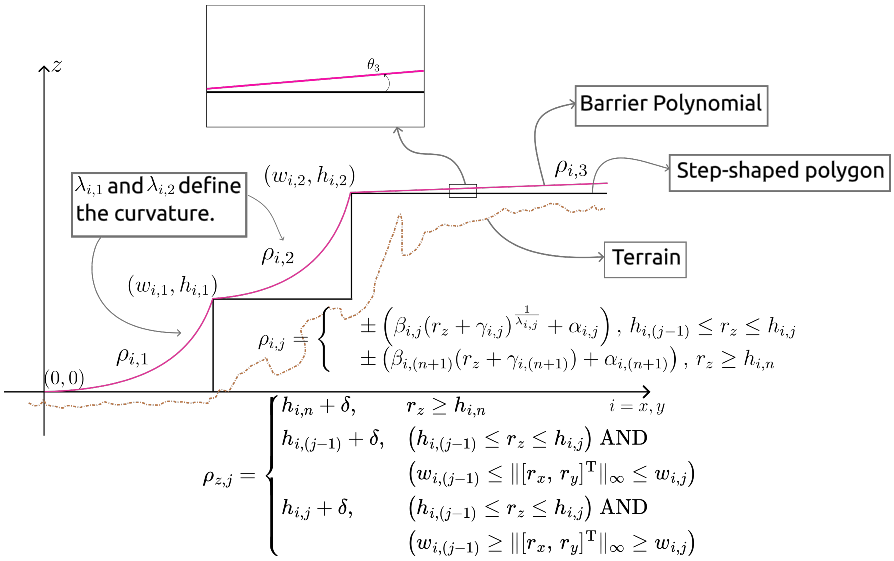


Figure 2.2: Illustration of barriers with $n = 2$.

demands. A higher degree polynomial will closely follow the step shape. Allowing for more freedom of movement for the lander. It is further considered that the $(n+1)^{\text{th}}$ barrier to be a linear polynomial instead of any higher order polynomial.

$$\rho_{i,j} = \begin{cases} \pm\left(\beta_{i,j}(r_z+\gamma_{i,j})^{\frac{1}{\lambda_{i,j}}}+\alpha_{i,j}\right), \ h_{i,(j-1)} \le r_z \le h_{i,j} \\ \pm\left(\beta_{i,(n+1)}(r_z+\gamma_{i,(n+1)})+\alpha_{i,(n+1)}\right), \ r_z \ge h_{i,n} \end{cases} \tag{2.5}$$

where $i = x, \ y$ and $j = 1, \ \ldots, \ n$. From the first barrier to the n^{th} barriers, the constants are defined as:

$$\alpha_{i,j} = w_{i,(j-1)}; \ \ \beta_{i,j} = \frac{w_{i,j} - w_{i,(j-1)}}{\left(h_{i,j} - h_{i,(j-1)}\right)^{\frac{1}{\lambda_{i,j}}}}; \ \ \gamma_{i,j} = -h_{i,(j-1)} \tag{2.6}$$

and $\lambda_{i,j}$ is a positive, even natural number. Further note that, $h_{i,0} = 0$, and $w_{i,0} = 0$, that is, origin is the landing site. For the $(n+1)^{\text{th}}$ barrier, slope angle of the barrier, $\theta_{(n+1)}$ with respect to the axis under consideration is chosen. The angle can be chosen as a small value (approx. $0.05° - 0.1°$) or a high value for a relatively flat plateau or a hill, respectively, beyond the canyon containing the lading site. The constants can now be defined as:

$$\alpha_{i,(n+1)} = w_{i,n}; \;\; \gamma_{i,(n+1)} = -h_{i,n}; \;\; \beta_{i,(n+1)} = \tan\left(\frac{\pi}{2} - \theta_{(n+1)}\right) \tag{2.7}$$

2.2.2 Barriers for vertical motion

It is imperative to define barriers at specific heights to avoid crash landing into the terrain due to vertical motion by including a small margin δ, to the height of the next lower barrier. The designer can choose this margin of safety as per requirement. However, if the lander is within the lateral bound of j^{th} step, but is above the height of $(j+1)^{\text{th}}$ step, the lander would keep on bouncing off the vertical barrier corresponding to the $(j+1)^{\text{th}}$ step. To address this issue, the vertical barriers are selected using the following simple comparison:

$$\rho_{z,j} = \begin{cases} h_{i,n} + \delta, & r_z \geq h_{i,n} \\ h_{i,(j-1)} + \delta, & \left(h_{i,(j-1)} \leq r_z \leq h_{i,j}\right) \text{ AND} \\ & \left(w_{i,(j-1)} \leq \|[r_x,\, r_y]^{\mathrm{T}}\|_\infty \leq w_{i,j}\right) \\ h_{i,j} + \delta, & \left(h_{i,(j-1)} \leq r_z \leq h_{i,j}\right) \text{ AND} \\ & \left(w_{i,(j-1)} \geq \|[r_x,\, r_y]^{\mathrm{T}}\|_\infty \geq w_{i,j}\right) \end{cases} \tag{2.8}$$

where, $j = 1, \ldots, n$.

2.3 DEVELOPMENT OF FEEDBACK GUIDANCE

The primary goal of this chapter is to develop a guidance law which uses the information about the barrier functions to generate necessary divert manoeuvre commands if the spacecraft moves towards the terrain. To this end, in this section, a novel feedback guidance law is developed, which can navigate the terrain and land safely and softly at the desired landing site consuming least amount of fuel.

2.3.1 Performance index for collision avoidance

To avoid crashing into the local terrain, a novel augmentation to the performance index is introduced as

$$J = \frac{1}{2}\int_0^{t_f}\left[\mathbf{a}^{\mathrm{T}}\mathbf{a} - \sum_i l_{3,i}e^{-\psi_i}\right]d\tau \tag{2.9}$$

where, $i = x, y, z$, $l_{3,i} > 0$ is a constant, and $e^{-\psi_i}$ is the augmentation term, where ψ_i is defined as:

$$\psi_i = \frac{l_{2,i}}{(d_i^2 + l_{1,i})} \tag{2.10}$$

where, $d_i \triangleq r_i - \rho_{i,j}$. Physically, d_i represents how far the lander is with respect to the barrier surface. When the lander is far away from the barriers, the augmentation is large and positive, so the term inside the integration in (2.9) is small. When the lander is close to the barrier, the augmentation is almost zero, increasing the cost, thus generating an acceleration command in direction opposite to the direction of motion to avoid crashing into terrain.

2.3.2 Guidance law

With the performance index in (2.9) subject to the equations of motion in (2.1) with $\mathbf{a}_{\mathrm{p}} = 0$, the Hamiltonian is written as:

$$H = \frac{1}{2}\left[\mathbf{a}^{\mathrm{T}}\mathbf{a} - \sum_i l_{3,i}e^{-\psi_i}\right] + \mathbf{p}_r^{\mathrm{T}}\mathbf{v} + \mathbf{p}_v^{\mathrm{T}}(\mathbf{a}+\mathbf{g}). \tag{2.11}$$

Since (2.11) is not linear in $\mathbf{a}$, the optimal control is of the form:

$$\frac{\partial H}{\partial \mathbf{a}} = 0 \Rightarrow \mathbf{a} = -\mathbf{p}_v. \tag{2.12}$$

For the equations of motion (2.1) and ψ_i from (2.10), the dynamics of the co-states are:

$$\dot{\mathbf{p}}_r = \frac{\partial}{\partial \mathbf{r}}\left[\frac{1}{2}\sum_i l_{3,i}e^{-\psi_i}\right] \tag{2.13}$$

$$\dot{\mathbf{p}}_v = -\mathbf{p}_r \tag{2.14}$$

From (2.13), defining $\mathbf{p} \triangleq [\dot{p}_{r_x},\ \dot{p}_{r_y},\ \dot{p}_{r_z}]$ where:

$$\dot{p}_{r_i} = l_{2,i} l_{3,i} \frac{d_i \mathrm{e}^{-\psi_i}}{\left(d_i^2 + l_{1,i}\right)^2}. \tag{2.15}$$

Moving forward, note that

$$\ddot{p}_{ri} = \frac{l_{2,i} l_{3,i}}{(d_i^2 + l_{1,i})^2} \left[1 - \frac{4d_i^2}{(d_i^2 + l_{1,i})} + \frac{2l_{2,i} d_i^2}{(d_i^2 + l_{1,i})^2}\right] e^{-\psi_i} v_x, \tag{2.16}$$

and by suitably choosing $l_{1,i}$ and $l_{2,i}$, $\operatorname{sgn} \ddot{p}_{ri} = -\operatorname{sgn} v_i$. Note that in horizontal motion, the lander moving close to the barriers as defined in Section 2.2.1 can follow either of the two cases: (1) $r_i > 0$ and $v_i > 0$, or (2) $r_i < 0$ and $v_i < 0$. In the first case, from (2.16) and (2.13), $\dot{p}_{ri} < 0$ and $\ddot{p}_{ri} \leq 0$, which implies that $\dot{p}_{ri}$ decreases, and thus, $p_{ri} \geq p_{rfi} - t_{\text{go}} \dot{p}_{ri}$. Then, from (2.12) and (2.14), in the first case,

$$a_i \leq -p_{vfi} - p_{rfi} t_{go} + \frac{t_{\text{go}}^2}{2} \dot{p}_{ri}. \tag{2.17}$$

On the other hand, in the second case, $\dot{p}_{ri} > 0$ and $\ddot{p}_{ri} \geq 0$ implying that $\dot{p}_{ri}$ increases, and thus, $p_{ri} \leq p_{rfi} - t_{\text{go}} \dot{p}_{ri}$. Hence, in the second case,

$$a_i \geq -p_{vfi} - p_{rfi} t_{\text{go}} + \frac{t_{\text{go}}^2}{2} \dot{p}_{ri}. \tag{2.18}$$

For obtaining a simplified closed form expression of $\mathbf{a}$ by simplifying (2.12)-(2.14), an approximation $p_{ri} \approx p_{rfi} - t_{\text{go}} \dot{p}_{ri}$ is considered in this paper. Similarly, $v_z < 0$ implies $p_{rz} \leq p_{rfz} - t_{\text{go}} \dot{p}_{rz}$. Following a similar analysis $a_z \geq -p_{vfz} - p_{rfz} t_{\text{go}} + \dot{p}_{rz}(t_{\text{go}}^2/2)$ is obtained, which leads to a sub-optimal guidance command as,

$$\mathbf{a} = -\mathbf{p}_{\text{vf}} - \mathbf{p}_{\text{rf}} t_{\text{go}} + \mathbf{p} \frac{t_{\text{go}}^2}{2}. \tag{2.19}$$

By substituting (2.19) in the equations of motion and integrating, expressions for position and velocity can be obtained as,

$$\mathbf{r} = \mathbf{r}_f - \mathbf{v}_f t_{\text{go}} - (\mathbf{p}_{\text{vf}} - \mathbf{g}) \frac{t_{\text{go}}^2}{2} - \mathbf{p}_{\text{rf}} \frac{t_{\text{go}}^3}{6} + \mathbf{p} \frac{t_{\text{go}}^4}{24}, \tag{2.20}$$

$$\mathbf{v} = \mathbf{v}_f + (\mathbf{p_{vf}} - \mathbf{g})\, t_{\text{go}} + \mathbf{p}_{\text{rf}} \frac{t_{\text{go}}^2}{2} - \mathbf{p} \frac{t_{\text{go}}^3}{6}. \tag{2.21}$$

Solving for terminal co-states from (2.20) and (2.21):

$$\mathbf{p}_{\text{rf}} = \frac{12}{t_{\text{go}}^3} \left[\left(\mathbf{r} - \mathbf{r}_f\right) + \left(\mathbf{v} + \mathbf{v}_f\right) \frac{t_{\text{go}}}{2} + \mathbf{p} \frac{t_{\text{go}}^4}{24} \right], \tag{2.22}$$

$$\mathbf{p}_{\text{vf}} = -\frac{6}{t_{\text{go}}^2} \left[\left(\mathbf{r} - \mathbf{r}_f\right) + \left(\mathbf{v} + 2\mathbf{v}_f\right) \frac{t_{\text{go}}}{3} + \mathbf{p} \frac{t_{\text{go}}^4}{72} \right] + \mathbf{g}. \tag{2.23}$$

Finally, substituting (2.22) and (2.23) in (2.19), the optimal terrain avoiding landing guidance (OTALG) law is obtained as:

$$\mathbf{a}_{\text{OTALG}} = \frac{6}{t_{\text{go}}^2} \mathbf{ZEM}(t) - \frac{2}{t_{\text{go}}} \mathbf{ZEV}(t) + \mathbf{p} \frac{t_{\text{go}}^2}{12}. \tag{2.24}$$

Comparing (2.4) and (2.24), note that the term $\mathbf{p}\frac{t_{\text{go}}^2}{12}$ is responsible for the collision avoidance divert manoeuvre.

2.3.3 Discussions

Define a vector $\mathbf{\Gamma} \triangleq \left[\dot{p}_{rx} \frac{t_{\text{go}}^2}{12},\ \dot{p}_{ry} \frac{t_{\text{go}}^2}{12},\ \dot{p}_{rz} \frac{t_{\text{go}}^2}{12} \right]^{\text{T}}$, which is the collision avoidance term in (2.24). Taking partial derivative of Γ_i w.r.t. r_i,

$$\frac{\partial \Gamma_i}{\partial r_i} = \frac{l_{2,i} l_{3,i} t_{\text{go}}^2}{12(d_i^2 + l_{1,i})^2} \left[1 - \frac{4d_i^2}{(d_i^2 + l_{1,i})} + \frac{2l_{2,i} d_i^2}{(d_i^2 + l_{1,i})^2} \right] e^{-\psi_i}. \tag{2.25}$$

The non-trivial critical points of (2.25) are when $e^{-\psi_i} = 0$, or $1 - \frac{4d_i^2}{(d_i^2 + l_{1,i})} + \frac{2l_{2,i} d_i^2}{(d_i^2 + l_{1,i})^2} = 0$. Considering the structure of ψ_i, the former does not have real solutions, so only the latter condition needs to be checked to determine the critical points. Simplifying the latter condition, gives the critical values of d_i as:

$$d_i^* = \pm \frac{\sqrt{\sqrt{l_{2,i}^2 - 2l_{1,i} l_{2,i} + 4l_{1,i}^2} + l_{2,i} - l_{1,i}}}{\sqrt{3}}. \tag{2.26}$$

The maximum magnitude of divert acceleration occurs at $d_i = d_i^*$, which is the critical distance of the barrier to the lander. The safety margin δ must be chosen to be greater than d_i^*. The safety margin can then be tuned according to (2.26) by setting $l_{1,i}$, and $l_{2,i}$

suitably. However, the safety margin of any step should be less than its height to prevent a lower level step interfering with the divert manoeuvre caused by a higher level step.

From a design perspective, knowledge of the maximum thrust the guidance law needs is a prerequisite. The performance index (2.9) is analysed to determine $T_{i_{\max}}$ where $i = x,\ y,\ z$. To ensure $J \geq 0$, the following must hold:

$$a_{i_{\max}}^2 - l_{3,i}\mathrm{e}^{-\psi_i} \geq 0 \Rightarrow |a_{i_{\max}}| \geq \sqrt{l_{3,i}\mathrm{e}^{-\psi_i}}. \tag{2.27}$$

From (2.27):

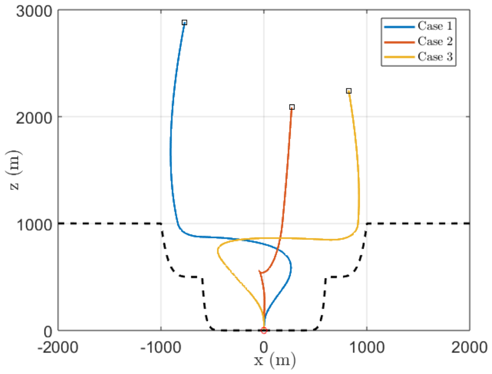
$$|T_{i_{\max}}| \geq m_0\sqrt{l_{3,i}\mathrm{e}^{-\psi_i}}. \tag{2.28}$$

From (2.15) and (2.24) observe that $l_{2,i}$ and $l_{3,i}$ have a direct affect on the maximum magnitude of divert acceleration. Thus, increasing $l_{2,i}$ and $l_{3,i}$ require a larger thrust to be generated by the lander, and therefore require a larger $|T_{i_{\max}}|$.

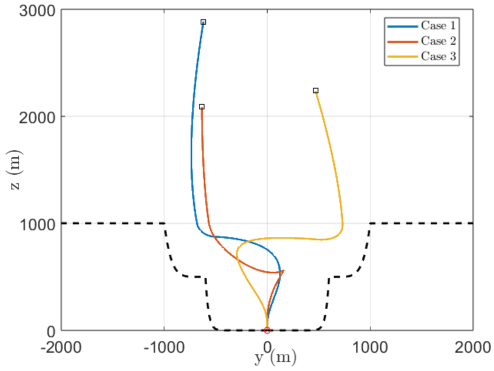
From (2.5), (2.8), Fig. 2.2 and the definition of d_i, $\operatorname{sgn} r_i = -\operatorname{sgn} d_i$. To avoid terrain successfully $|d_i|$ should increase which implies, from (2.15), $\operatorname{sgn} p_i = \operatorname{sgn} d_i$ and therefore the product $l_{2,i}l_{3,i} > 0$. However, from (2.27), $l_{3,i} > 0$ which implies $l_{2,i} > 0$ as well.

2.4 SIMULATION RESULTS

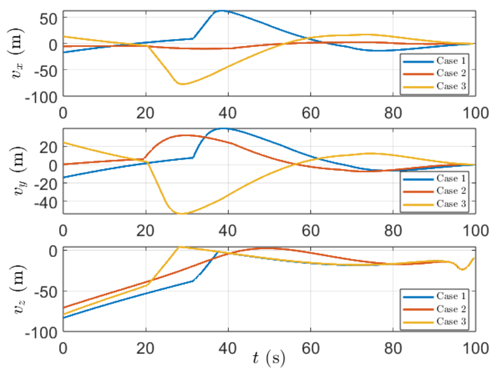
To demonstrate the effectiveness of the proposed guidance law, results from simulations are presented in this section. A simplified version of simulation parameters of [1] have been considered for simulation. A point-mass lander with 3DoF is assumed, with specific impulse $I_{\text{sp}} = 225$ s and $T_{\max} = 31000$ N (equivalent to ten thrusters, each with 3100 N max thrust). The desired terminal states are $\mathbf{r}_f^d = [0,\ 0,\ 0]^{\mathrm{T}}$ m, $\mathbf{v}_f^d = [0,\ 0,\ 0]^{\mathrm{T}}$ m/s, at terminal time $t_f = 100$ s. The simulation is stopped when $r_z = 0.05$ m or terminal time is achieved, whichever occurs first. To emulate a trench surrounding the landing site on Mars, consider a terrain that can be modelled as a 2-step, flat-top shape (as shown in Fig, 2.2). The height and width from the origin of each step are given in Table 2.1.



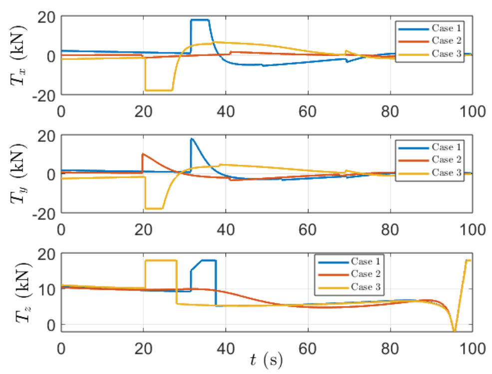
(a) Trajectories in x-z plane.



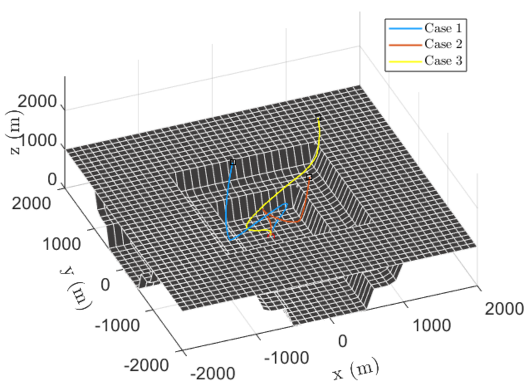
(b) Trajectories in y-z plane.



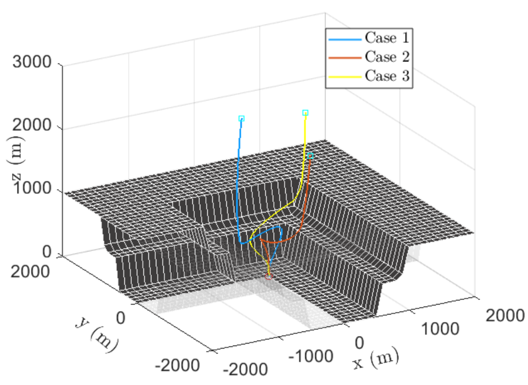
(c) Velocity profiles



(d) Thrust profiles.



(e) Trajectories in 3D representation - 1.



(f) Trajectories in 3D representation - 2.

Figure 2.3: Trajectories, velocity and thrust profile with no perturbations.

To design the barriers (see (2.5) and (2.8)), $\theta_3 = 0.05°$, with $\lambda_{i,2} = 6$, and $\lambda_{i,1} = 20$ are chosen. Further, the margin of safety in the vertical barrier is $\delta = 1.2d_i^*$. The guidance law constant are chosen as $l_{1,i} = 1$, $l_{2,i} = 3000$, and $l_{3,i} = 280$. Thus, from (2.26), $\delta = 53.67$ m. Finally, the local gravity at Mars is assumed to be $g = [0,\ 0,\ -3.7114]^{\mathrm{T}}$ m/s^2, and acceleration due to gravity on Earth $g_e = 9.807$ m/s^2.

Table 2.1: Model for terrain approximation

	Height (m)	Width (m)
Step-1	500	600
Step-2	1000	1000

2.4.1 Terrain collision avoidance

To portray the effectiveness of OTALG in terms of terrain avoidance, the trajectories of $N = 3$ sample initial conditions are presented. Trajectories corresponding to these initial conditions are shown in Fig. 2.3a, Fig. 2.3b, Fig. 2.3e and Fig. 2.3f. The velocity profiles of the three cases are shown in Fig. 2.3c, and the thrust profiles are shown in Fig. 2.3d.

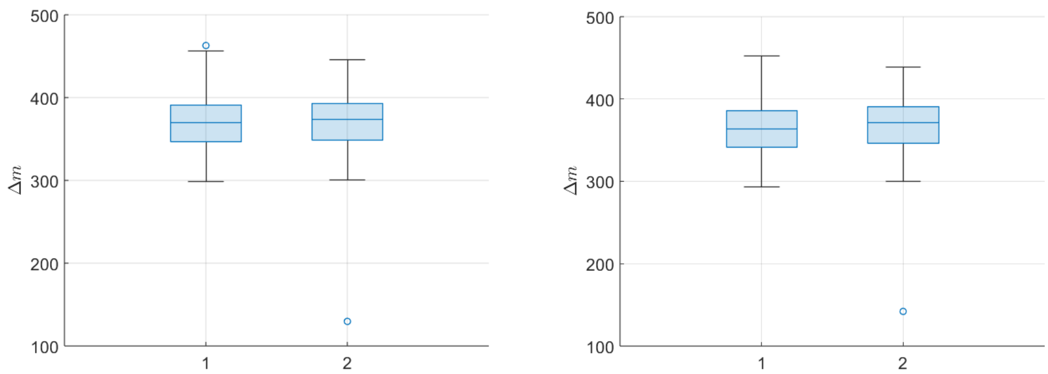
It is clear from the velocity profile in Fig. 2.3c that the guidance law brings the lander softly to the landing site. There are no sudden jerks in either the trajectories or the velocity profiles. From the trajectory figures and the velocity profiles the guidance law working to avoid the terrain can be observed. Consider Case 3 in Fig. 2.3a. When the lander is near the terrain, the acceleration command generated, as shown in Fig. 2.3d, by the collision avoidance term pushes it away from the terrain by changing the velocity

Table 2.2: Sample Initial Conditions for Illustration

Case #	$\mathbf{r}_0$	$\mathbf{v}_0$	m_0
Case 1	$[-769.42, -619.63, 2883.33]^{\mathrm{T}}$	$[-16.78, -14.08, -83.36]^{\mathrm{T}}$	1961.80
Case 2	$[269.35, -634.30, 2086.65]^{\mathrm{T}}$	$[-4.98, 0.29, -70.89]^{\mathrm{T}}$	1916.55
Case 3	$[823.91, 467.70, 2240.03]^{\mathrm{T}}$	$[13.81, 24.28, -79.47]^{\mathrm{T}}$	1959.43

Table 2.3: Initial Conditions Distribution Setup

Initial Condition	μ	σ^2
x_0 (m)	0	350
y_0 (m)	0	350
z_0 (m)	2500	500
v_{x0} (m/s)	0	10
v_{y0} (m/s)	0	10
v_{z0} (m/s)	-80	10
m_0 (kg)	1900, 2100	100



(a) Fuel consumption statistics under no perturbations.

(b) Fuel consumption statistics under atmospheric perturbations.

Figure 2.4: Fuel consumption statistics.

direction, as clearly observed in the v_x plot of Fig. 2.3c. In all three cases as the lander approaches landing site, the proposed guidance law generates the acceleration in the direction opposite to the direction of motion as evident from Fig. 2.3d, and softly reaches the landing site.

2.4.2 Fuel consumption

To compare the efficacy of OTALG (Algorithm 1) in terms of fuel consumption, comparative studies are performed with the work done in [43] (Algorithm 2), which was shown to be near-fuel optimal with respect to the classical ZEM/ZEV guidance law in [10]. Further, the performance is also tested under bounded atmospheric perturbations modelled as

$$\mathbf{a}_p(t) = 0.5\mathbf{a}_{\text{OTALG}} \sin\left(\frac{\pi}{3}t\right). \tag{2.29}$$

where, $\mathbf{a}_{\text{OTALG}}$ is as obtained in (2.24). Paired t-test is performed between the net fuel consumption of the two guidance laws. In this test, there are $N = 300$ data points for fuel consumption, initial conditions for which are taken from a normally distributed set in Table 2.3. The null hypothesis for the test is $H_0 : d \triangleq \Delta m_1 - \Delta m_2 = 0$, where Δm_j is the fuel consumption for Algorithm j. For 0.05 significance level, $t_{\text{crit}} = 1.9679$, $\bar{d} = -0.3581$ and $\sigma_{\bar{d}} = 1.2719$. Thus, for the paired t-test, $t = -0.2815$ implying $|t| < t_{\text{crit}}$. From the box chart in Fig. 2.4a, the median fuel consumption is marginally smaller for the proposed law. Thus, it can be inferred that the proposed guidance law w.r.t. the guidance law in [43] are statistically almost similar in terms of near-fuel optimal performance under ideal scenario. Next, the paired t-test on the data-set generated with the same initial conditions under atmospheric perturbations as defined in (2.29) is performed, which gives $\bar{d} = -4.2582$ and $\sigma_{\bar{d}} = 1.2116$. Thus, $t = -3.5415$ implying that $|t| > t_{\text{crit}}$. Moreover, from the box charts in Figure 2.4b for the net fuel consumption under atmospheric perturbations, it is observed that the median fuel consumption is lesser in Algorithm 1, implying that under bounded atmospheric perturbations, the proposed guidance law achieves far superior performance in terms of near-fuel optimality when compared to Algorithm 2.

2.5 CONCLUSIONS

Optimal guidance law for fuel-optimal, hazardous terrain landing, named OTALG, has been presented in this chapter. The terrain has been modelled as n-step-shaped polygons, and barriers have been defined over the terrain model using piecewise-smooth polynomials. Considering terrain avoidance as a penalty on the running cost, fuel optimal guidance law has been derived. The efficacy of the guidance law is tested in a trench in the Martian environment, emulated using a flat-top, 2-step shape. The terrain avoidance guidance successfully avoids the terrain. Using extensive simulations, the near-fuel optimality is confirmed as well. The guidance law avoids terrain under bounded perturbation and maintains near-fuel optimality. While the success of OTALG in terms of near-fuel optimality, terrain avoidance and soft landing is promising, the accuracy of precision soft landing remains poor, especially under disturbances and uncertainties. To that end, the proposed guidance law must be expanded to include a globally stabilising controller to improve the accuracy of precision soft landing and make the guidance law robust against disturbances, such as those caused by winds, and uncertainties, such as those caused by thrust lag and noises.

CHAPTER 3

ROBUST OTALG USING MULTIPLE SLIDING SURFACES

The results in the previous chapter showed good performance in terms of near-fuel-optimality and terrain avoidance compared to the existing literature. And while OTALG is able to avoid the terrain with near-fuel optimality, it cannot guarantee robustness against external perturbations. In this chapter, the OTALG is augmented with sliding mode control (SMC) to improve robustness and the accuracy of the precision soft landing. To this end, multiple sliding surfaces (MSS) are used here. The first sliding surface is established to monitor the positional error relative to the target landing site, with a virtual controller introduced to ensure convergence of this sliding variable, while the second sliding surface is implemented to ensure that the first sliding variable follows the virtual control.

3.1 USEFUL LEMMAS AND NOTATION

Lemma 3.1. *(Young's Inequality) For any vector* $\mathbf{x}$*,* $\mathbf{y} \in \mathbb{R}^n$*,* $\mathbf{x}^{\mathrm{T}}\mathbf{y} \leq \frac{\|\mathbf{x}\|^a}{a} + \frac{\|\mathbf{y}\|^b}{b}$ *holds true where* $a,\ b > 1$ *and* $(a-1)(b-1) = 1$*.*

Lemma 3.2. *[22; 31] Consider the system,*

$$\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}(t)),\ \mathbf{x}(t_0) = \mathbf{x}_0. \tag{3.1}$$

Suppose there exists a Lyapunov function $V(\mathbf{x})$ *such that,*

$$\dot{V}(\mathbf{x}) \leq -\left(\alpha V^l(\mathbf{x}) + \beta V^m(\mathbf{x})\right)^k + \eta \tag{3.2}$$

where α, β, l, m, $k > 0$, $lk < 1$, $mk > 1$ and $0 < \eta < \infty$. Then the trajectories of (3.1) *are practically fixed-time stable, with residual set*

$$\left\{ \lim_{t \to T} \mathbf{x} \,|\, V(\mathbf{x}) \leq \min \left\{ \begin{array}{c} \alpha^{-\frac{1}{l}} \left(\frac{\eta}{1-\theta^k}\right)^{\frac{1}{lk}}, \\ \beta^{-\frac{1}{l}} \left(\frac{\eta}{1-\theta^k}\right)^{\frac{1}{mk}} \end{array} \right\} \right\} \tag{3.3}$$

where $\theta \in (0, 1]$ and the settling time, T is upper bounded by

$$T \leq \left(\frac{1}{\alpha^k \theta^k (1 - lk)} + \frac{1}{\beta^k \theta^k (mk - 1)} \right). \tag{3.4}$$

Definition 3.1 (Signum function). The signum function, $\mathrm{sgn}(\cdot)$ is defined as:

$$\mathrm{sgn}\,\Delta \triangleq \begin{cases} -1 & \text{if } \Delta < 0 \\ 0 & \text{if } \Delta = 0 \\ 1 & \text{if } \Delta > 0 \end{cases}. \tag{3.5}$$

Further, consider $\mathbf{x} \in \mathbb{R}^n$, then

$$\mathrm{sgn}\,\mathbf{x} \triangleq [\mathrm{sgn}\,x_1,\ \mathrm{sgn}\,x_2, \ldots, \mathrm{sgn}\,x_n]^{\mathrm{T}}. \tag{3.6}$$

3.2 MSS DESIGN

3.2.1 Surface 1

The first sliding surface is defined as $\mathbf{s}_1 \triangleq \mathbf{r} - \mathbf{r}_f^d$ where $\mathbf{r}_f^d$ is the desired terminal position. This sliding surface is used to track the position error. To track $\mathbf{s}_1$ and drive the position error to zero, a virtual controller is defined as $\dot{\mathbf{s}}_1 = -\frac{\Lambda}{t_{\mathrm{go}}}\mathbf{s}_1$ where $\Lambda > 0$ is constant.

Theorem 3.3. The virtual controller defined by $\dot{\mathbf{s}}_1 = -\frac{\Lambda}{t_{\mathrm{go}}}\mathbf{s}_1$, is globally stable. Further, both $\mathbf{s}_1$ and its derivative, under the reaching law defined by the virtual controller reach zero in finite time.

Proof. Global stability can be guaranteed for the virtual controller using Lyapunov's second method, and choosing the candidate Lyapunov function as $V_1 = 0.5\mathbf{s}_1^{\mathrm{T}}\mathbf{s}_1$. From direct observation, it is clear that V_1 is positive definite everywhere, except at $\mathbf{s}_1 = \mathbf{0}$ where

$V_1(\mathbf{s}_1 = \mathbf{0}) = 0$. Further, the candidate Lyapunov function is radially unbounded. To guarantee global stability, the time derivative of V_1 must be negative definite everywhere, that is $\dot{V}_1 = \mathbf{s}_1^{\mathrm{T}}\dot{\mathbf{s}}_1 < 0$. This implies:

$$\dot{V}_1 = -\frac{\Lambda}{t_{\text{go}}}\mathbf{s}_1^{\mathrm{T}}\mathbf{s}_1 < 0 \tag{3.7}$$

Equation (3.7) proves that the virtual controller is globally stable, and concludes the first part of the proof. To analyse the finite-time stability, the differential equation given by the virtual controller is solved component-wise.

$$\begin{aligned}
&\frac{\mathrm{d}s_{1i}}{\mathrm{dt}} = -\frac{\Lambda}{t_{\text{go}}}s_{1i} \\
&\Rightarrow \ln s_{1i} = \Lambda \ln t_{\text{go}} + C_1 \\
&\Rightarrow \begin{aligned} s_{1i} &= s_{1i0}\left(\tfrac{t_{\text{go}}}{t_f}\right)^{\Lambda} \\ \dot{s_{1i}} &= -\tfrac{\Lambda}{t_f}s_{1i0}\left(\tfrac{t_{\text{go}}}{t_f}\right)^{\Lambda-1} \end{aligned}
\end{aligned} \tag{3.8}$$

The sliding surface and its derivative can be made finite-time convergent at $t = t_f$ by setting $\Lambda > 1$, and thus concluding the proof of this theorem. ■

3.2.2 Surface 2

At the beginning of powered descent stage, due to a variety of reasons, such as disturbances caused due to model uncertainty and atmospheric perturbations, the relation $\dot{\mathbf{s}}_1 = -\frac{\Lambda}{t_{\text{go}}}\mathbf{s}_1$ is not satisfied. The guidance law should be designed to drive $\dot{\mathbf{s}}_1$ from its initial condition to $\dot{\mathbf{s}}_1 = -\frac{\Lambda}{t_{\text{go}}}\mathbf{s}_1$, maintain it there regardless of any disturbances, and eventually drive $\mathbf{s}_1$ to zero. To achieve this, a second sliding surface is proposed as:

$$\mathbf{s}_2 = \dot{\mathbf{s}}_1 + \frac{\Lambda}{t_{\text{go}}}\mathbf{s}_1 \tag{3.9}$$

From $\mathbf{s}_1$ and its dynamics, $\dot{\mathbf{s}}_1 = \mathbf{v} - \mathbf{v}_f^d$ and $\ddot{\mathbf{s}}_1 = \mathbf{a}_c + \mathbf{g} - \dot{\mathbf{v}}_f^d$. Taking time derivative of (3.9):

$$\dot{\mathbf{s}}_2 = \ddot{\mathbf{s}}_1 + \frac{\Lambda}{t_{\text{go}}}\dot{\mathbf{s}}_1 + \frac{\Lambda}{t_{\text{go}}^2}\mathbf{s}_1$$

$$\Rightarrow \dot{\mathbf{s}}_2 = \mathbf{a}_c + \mathbf{g} - \dot{\mathbf{v}}_f + \frac{\Lambda}{t_{\text{go}}}\left(\mathbf{v} - \mathbf{v}_f^d\right) + \frac{\Lambda}{t_{\text{go}}^2}\left(\mathbf{r} - \mathbf{r}_f^d\right) \tag{3.10}$$

Since $\dot{\mathbf{s}}_2$, as shown in (3.10), has the acceleration term, the relative degree of $\mathbf{s}_2$ is 1, an appropriately chosen guidance law can drive $\mathbf{s}_2$ to zero.

Theorem 3.4. *Consider the system dynamics given by* (2.1) *and the sliding surface* $\mathbf{s}_2$ *given by* (3.9)*, the guidance law:*

$$\mathbf{a}_c = \mathbf{a}_{\text{OTALG}} - \mathbf{\Phi}\,\text{sgn}\,\mathbf{s}_2 - \mathbf{g} \tag{3.11}$$

where $\mathbf{a}_{\text{OTALG}}$ *is given by* (2.24)*, will drive the second sliding surface* $\mathbf{s}_2$ *to zero, where* $\mathbf{\Phi} = \text{diag}\left\{\Phi_x,\ \Phi_y,\ \Phi_z\right\}$ *is the sliding parameter, which depends on the distance of the lander to the barriers* d_i *and time-to-go* t_{go}.

Proof. The candidate Lyapunov function is chosen as $V_2 = 0.5\mathbf{s}_2^{\text{T}}\mathbf{s}_2$. From direct observation, V_2 is positive definite everywhere except at origin where it is zero and is radially unbounded everywhere in the domain of $\mathbf{s}_2$ that is $\mathbb{R}^3$. To prove global stability, the negative definiteness of the time derivative of V_2 is analysed as:

$$\begin{aligned}
&\dot{V}_2 = \mathbf{s}_2^{\text{T}}\dot{\mathbf{s}}_2 \\
&\Rightarrow \dot{V}_2 = \mathbf{s}_2^{\text{T}}\left(\mathbf{a}_{\text{OTALG}} - \mathbf{\Phi}\,\text{sgn}\,\mathbf{s}_2 - \dot{\mathbf{v}}_f + \frac{\Lambda}{t_{\text{go}}}\left(\mathbf{v} - \mathbf{v}_f^d\right) + \frac{\Lambda}{t_{\text{go}}^2}\left(\mathbf{r} - \mathbf{r}_f^d\right)\right).
\end{aligned} \tag{3.12}$$

From the nature of the landing guidance problem considered here, $\mathbf{r}_f^d = \mathbf{0}$ and $\mathbf{v}_f^d = \mathbf{0}$, and consequently $\dot{\mathbf{v}}_f^d = \mathbf{0}$. With these considerations, from (3.9):

$$\mathbf{s}_2 = \mathbf{v} + \frac{\Lambda}{t_{\text{go}}}\mathbf{r}. \tag{3.13}$$

Then, from (2.2), (2.24) and (3.12):

$$\dot{V}_2 = \mathbf{s}_2^{\text{T}}\left[\left(\frac{\Lambda - 6}{t_{\text{go}}^2}\mathbf{r} + \frac{\Lambda - 4}{t_{\text{go}}}\mathbf{v}\right) + \mathbf{p}\frac{t_{\text{go}^2}}{12} - \mathbf{\Phi}\,\text{sgn}\,\mathbf{s}_2\right]. \tag{3.14}$$

For $\Lambda = 2,\ 3$, from (3.13) and (3.14):

$$\dot{V}_2 = \left(\frac{\Lambda - 4}{t_{\mathrm{go}}}\right)\mathbf{s}_2^{\mathrm{T}}\mathbf{s}_2 + \mathbf{s}_2^{\mathrm{T}}\left(\mathbf{p}\frac{t_{\mathrm{go}}^2}{12} - \mathbf{\Phi}\,\mathrm{sgn}\,\mathbf{s}_2\right). \tag{3.15}$$

Observe that the first term in (3.15), for $\Lambda = 2,\ 3$, is negative. To ensure the second term is negative as well, $\mathbf{\Phi}$ must be chosen suitably. Examining the second term component-wise for negative semi-definiteness:

$$s_{2i}\left(p_i\frac{t_{\mathrm{go}}^2}{12} - \Phi_i\,\mathrm{sgn}\,s_{2i}\right) \leq 0$$

$$\Rightarrow s_{2i}\left(p_i\frac{t_{\mathrm{go}}^2}{12}\right) \leq \Phi_i|s_{2i}|. \tag{3.16}$$

To enforce the inequality in (3.16), each element of $\mathbf{\Phi}$ can be chosen to satisfy:

$$\Phi_i \geq |p_i|\frac{t_{\mathrm{go}}^2}{12}. \tag{3.17}$$

Thus, setting Φ_i according to the inequality (3.17) will make (3.15) negative definite and complete the proof for this theorem. ■

3.2.3 Robustness Analysis

Now, the disturbances in the environment, as expressed in (2.1), that can cause the spacecraft to deviate from its nominal trajectory are considered. Equation (3.14) now becomes:

$$\dot{V}_2 = \mathbf{s}_2^{\mathrm{T}}\left[\left(\frac{\Lambda - 6}{t_{\mathrm{go}^2}}\mathbf{r} + \frac{\Lambda - 4}{t_{\mathrm{go}^2}}\mathbf{v}\right) + \mathbf{a}_{\mathrm{p}} + \mathbf{p}\frac{t_{\mathrm{go}^2}}{12} - \mathbf{\Phi}\,\mathrm{sgn}\,\mathbf{s}_2\right]. \tag{3.18}$$

Using a similar analysis used in the proof for Theorem 3.4, the following condition for guaranteed robustness holds:

$$\Phi_i \geq \left|p_i\frac{t_{\mathrm{go}}^2}{12} + a_{\mathrm{p}_i}\right|. \tag{3.19}$$

The disturbances represented by $\mathbf{a}_{\mathrm{p}}$ are unknown, however, an upper bound is generally available i.e. $|a_{\mathrm{p}_i}| \leq a_{\mathrm{p_{MAX}}}$. Therefore, to determine the range of Φ_i for guaranteed robustness triangle inequality is used in (3.19), and substitute the upper bound for the unknown perturbations:

$$\begin{aligned}\Phi_i &\geq \left|p_i \frac{t_{\text{go}}^2}{12}\right| + \left|a_{\text{p}_i}\right| \\ \Rightarrow \Phi_i &\geq \left|p_i \frac{t_{\text{go}}^2}{12}\right| + a_{\text{pMAX}}.\end{aligned} \tag{3.20}$$

3.3 ON THE CHOICE OF SLIDING PARAMETER

A common practice in sliding mode control is to fix the sliding parameter to be constant based on *a-priori* obtained estimates of disturbances. However, the lower bound on Φ_i defined in (3.19) depends on the states and the t_{go}, and thus the sliding parameter should also be defined in the same manner. The maximum value of RHS in (3.19) can be determined and used as a sliding parameter, but it is an aggressive choice. An aggressive sliding parameter improves terminal precision as it commands a higher acceleration to turn towards the sliding surface as quickly as possible in the state-space and then maintain the states close to the sliding surfaces, requiring a higher control effort. Contrary to this, the states must come out of the vicinity of the sliding surface to avoid any collision with the terrain. To avoid aggressive sliding mode control and still achieve good accuracy in precision soft landing, the sliding parameter is proposed as:

$$\Phi_i = k_1 \left|p_i\right| \frac{t_{\text{go}}^2}{12} + k_2 a_{\text{pMAX}} \tag{3.21}$$

where, $k_1,\ k_2$ are tunable positive constants. Setting $k_1,\ k_2 = 1$ guarantees (3.19) is satisfied. However, this requires divert manoeuvre to be executed with thrust higher than what is actually necessary. Setting $k_1,\ k_2 < 1$ may, in fact, be sufficient to successfully execute the divert manoeuvre while still maintaining the robustness in precision soft landing. This, however, may lead to violations of (3.19), leading to loss of finite time stability and guaranteed robustness. We now show that even if the sliding constraint in (3.19) is violated, by choosing the sliding parameters as (3.21) with $k_1,\ k_2 < 1$, the guidance law in (3.11) can still maintain robustness in precision soft landing. To this end, the duration for which the dominant divert manoeuvre is active must be defined, and prove that the duration of the dominant divert manoeuvre is finite. Finally, it can be shown that with the sliding parameter chosen in (3.21), practical fixed time stability can be achieved.

Definition 3.2 (Duration of dominant divert manoeuvres). Dominant divert manoeuvre is said to begin when $|p_i|t_{\text{go}}^2/12 > |(6/t_{\text{go}}^2)\text{ZEM}_i - (2/t_{\text{go}})\text{ZEV}_i|$ and it ends when $|p_i|t_{\text{go}}^2/12 < |(6/t_{\text{go}}^2)\text{ZEM}_i - (2/t_{\text{go}})\text{ZEV}_i|$. Further, in some cases when only a small divert acceleration is required, the dominant divert manoeuvre starts even with $|p_i|t_{\text{go}}^2/12 < |(6/t_{\text{go}}^2)\text{ZEM}_i - (2/t_{\text{go}})\text{ZEV}_i|$ if $\operatorname{sgn}\dot{d}_i$ changes. In such cases, the dominant divert manoeuvre ends when $\operatorname{sgn}\dot{d}_i$ changes again.

Remark 1. From the analysis of p_i in Section 2.3.3, the behaviour of $|p_i|$ with respect to d_i, as shown in Fig. 3.1, is a decreasing function for all $|d_i| > |d_i^*|$ where d_i^* is the value of d_i for which p_i is maximum, that is, $|p_i(d_i^*)| = p_{\text{MAX}}$. Also, note from the divert term in (2.24) and its definition in (2.15), for $d_i > 0$:

$$\frac{\partial(p_i t_{\text{go}}^2/12)}{\partial d_i} = \Gamma\left[1 - \frac{4d_i^2}{(d_i^2 + l_{1,i})} + \frac{2l_{2,i}d_i^2}{(d_i^2 + l_{1,i})^2}\right]\frac{t_{\text{go}}^2}{12}.$$

where $\Gamma = \frac{l_{2,i}l_{3,i}e^{-\psi_i}}{(d_i^2+l_{1,i})^2} > 0$ and $\frac{t_{\text{go}}^2}{12} \geq 0$. The above expression can be then made negative for all values of $d_i > d_i^*$ by suitably choosing $l_{1,i}$ and $l_{2,i}$. This implies that when the lander approaches the barriers, that is, d_i decreases, with initial conditions $d_i > d_i^*$, the divert term will necessarily increase and therefore will cause the dominant divert manoeuvre to begin. Similar logic also holds true for $d_i < 0$.

Proposition 1. The duration of a dominant divert manoeuvre is finite.

Proof. Consider that the dominant divert manoeuvre begins for some $|d_i| > |d_i^*|$. During this time, the larger divert acceleration implies that the rate at which the lander approaches the barriers reduces. If the maximum thrust that the lander can generate is sufficiently large, then the sign of velocity, that is the direction of the velocity vector, will change and the lander will start to move away from the barrier. Observe that, in (2.24) when t_{go} is large and $|d_i| \gg |d_i^*|$, the magnitude of ZEM/ZEV component is of the order of

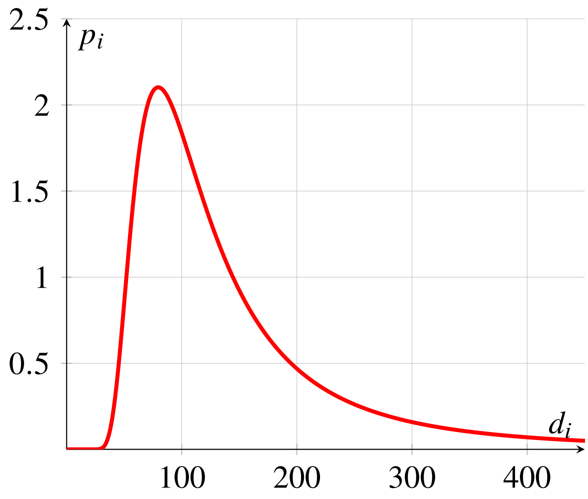


Figure 3.1: Behaviour of p_i with respect to d_i for $l_{1,i} = 1,\ l_{2,i} = 9500,\ l_{3,i} = 500$.

$O(1/t_{\text{go}} + g_i)$, and when the t_{go} is small, the magnitude of ZEM/ZEV component is of the order of $O(1/t_{\text{go}}^2)$. Finally, for the mid-ranges of t_{go}, the magnitude of ZEM/ZEV component has the order of $O(g_i)$. Since initially $|d_i| \gg |d_i^*|$, $|p_i|t_{\text{go}}^2/12 \approx 0$. However, as t increases, t_{go} decreases and $\lim_{t_{\text{go}}\to 0} |p_i|t_{\text{go}}^2/12 = 0$. This implies that at the end of the landing mission the magnitude of divert acceleration term in (2.24) comes sufficiently close to zero which is less than the magnitude of ZEM/ZEV term, and the dominant divert manoeuvre comes to an end in finite time. Since the mission is a fixed final time mission, this also implies that any dominant divert manoeuvre will come to an end in finite time. ■

Theorem 3.5. *The trajectory of the lander, governed by the dynamics given by* (2.1) *and under the guidance law* $\mathbf{a}_c = \mathbf{a}_{\text{OTALG}} - \mathbf{\Phi}\,\text{sgn}\,\mathbf{s}_2 - \mathbf{g}$ *as defined in* (3.11)*, have practical fixed-time stability (PFTS).*

Proof. From (3.15), $\dot{V}_2 = \sum \dot{V}_{2i},\ i = x,\ y,\ z$ where:

$$\dot{V}_{2i} = \left(\frac{\Lambda - 4}{t_{\text{go}}}\right) s_{2i}^2 + s_{2i}\left(p_i \frac{t_{\text{go}}^2}{12} - \Phi_i\,\text{sgn}\,s_{2i}\right)$$

$$\Rightarrow \dot{V}_{2i} = \left(\frac{\Lambda - 4}{t_{\text{go}}}\right) |s_{2i}|^2 - \Phi_i|s_{2i}| + s_{2i} p_i \frac{t_{\text{go}}^2}{12}. \tag{3.22}$$

For ease in simplification, in (3.22) consider A $= \left(\frac{\Lambda-4}{t_{\mathrm{go}}}\right)|s_{2i}|^2 - \Phi_i|s_{2i}|$ and B $= s_{2i}p_i\frac{t_{\mathrm{go}}^2}{12}$.

From A:

$$
\begin{aligned}
\mathrm{A} &= \left(\frac{\Lambda-4}{t_{\mathrm{go}}}\right)(2V_{2i}) - \Phi_i|s_{2i}| \\
\Rightarrow \mathrm{A} &\leq \left(\frac{\Lambda-4}{t_{\mathrm{go}}}\right)(2V_{2i}) - \Phi_i\left(\sqrt{2}V_{2i}^{1/2}\right) \\
\Rightarrow \mathrm{A} &\leq -\sqrt{2}\Phi_i V_{2i}^{1/2}. \qquad (3.23)
\end{aligned}
$$

From Lemma 3.1:

$$
\begin{aligned}
\mathrm{B} &= s_{2i}p_i\frac{t_{\mathrm{go}}^2}{12} \\
\Rightarrow \mathrm{B} &\leq \left(\frac{1}{1+p_1}|s_{2i}|^{1+p_1} + \frac{p_1}{1+p_1}|p_i|^{\frac{1+p_1}{p_1}}\right)\frac{t_{\mathrm{go}}^2}{12} \\
\Rightarrow \mathrm{B} &\leq \left(\frac{2^{\frac{1+p_1}{2}}}{1+p_1}V_{2i}^{\frac{1+p_1}{2}} + \frac{p_1}{1+p_1}|p_i|^{\frac{1+p_1}{p_1}}\right)\frac{t_{\mathrm{go}}^2}{12}. \qquad (3.24)
\end{aligned}
$$

For $\dot{V}_{2i}$, using (3.23) and (3.24):

$$
\dot{V}_{2i} \leq -\sqrt{2}\Phi_i V_{2i}^{1/2} + \left(\frac{2^{\frac{1+p_1}{2}}}{1+p_1}V_{2i}^{\frac{1+p_1}{2}} + \frac{p_1}{1+p_1}|p_i|^{\frac{1+p_1}{p_1}}\right)\frac{t_{\mathrm{go}}^2}{12} \qquad (3.25)
$$

$$
\dot{V}_{2i} \leq -\left(\sqrt{2}\Phi_i V_{2i}^{1/2} - \frac{2^{\frac{1+p_1}{2}}}{1+p_1}V_{2i}^{\frac{1+p_1}{2}}\frac{t_{\mathrm{go}}^2}{12}\right) + \frac{p_1}{1+p_1}|p_i|^{\frac{1+p_1}{p_1}}\frac{t_{\mathrm{go}}^2}{12} \qquad (3.26)
$$

where $p_1 > -1$. Using the results of Lemma 3.2, comparing (3.26) with (3.2) to get

$$
\begin{aligned}
\alpha_i &= \sqrt{2}\Phi_i \\
\beta_i &= -\frac{2^{\frac{1+p_1}{2}}}{1+p_1}\frac{t_{\mathrm{go}}^2}{12} \\
\eta_i &= \frac{p_1}{1+p_1}|p_i|^{\frac{1+p_1}{p_1}}\frac{t_{\mathrm{go}}^2}{12} \\
k = 1,\ l = 1/2,\ \ m &= \frac{1+p_1}{2}
\end{aligned} \qquad (3.27)
$$

where the conditions set by Lemma 3.2 enforce $p_1 > -1$. The settling time is given as:

$$
T_i \leq \frac{\sqrt{2}}{\Phi_i\theta} - \frac{\sqrt{2}(1+p_1)}{2^{\frac{p_1}{2}}(p_1-1)\theta}\frac{12}{t_{\mathrm{go}}^2} \qquad (3.28)
$$

where $i = x, y, z$. The RHS of (3.28) must be strictly positive and $\max T_i < t_f$, which lead to the following conditions:

$$\Phi_i \leq \frac{2^{\frac{p_1}{2}}(p_1-1)}{1+p_1}\frac{t_{go}^2}{12} = \text{L (say)} \tag{3.29}$$

$$\Phi_i \geq \frac{1}{\frac{1+p_1}{2^{\frac{p_1}{2}}(p_1-1)}\frac{12}{t_{go}^2} + \frac{t_f\theta}{\sqrt{2}}} = \text{M (say)}. \tag{3.30}$$

Substituting the proposed sliding parameter (3.21) in (3.29) and rearranging the terms with $|p_i| = p_{\text{MAX}}$ gives

$$k_1 p_{\text{MAX}} + 12k_2 a_{p_{\text{MAX}}}/t_{go}^2 \leq 2^{\frac{p_1}{2}}(p_1-1)/(1+p_1). \tag{3.31}$$

For all values of $t_{go} \in (0, t_f]$, there exists a $p_1 > 1$ that satisfies (3.31). Further, in the condition given by (3.30), observe that when t_{go} is large, $\Phi_i \geq M \approx \sqrt{2}/(t_f\theta)$, which can be satisfied by choosing a sufficiently large $a_{p_{\text{MAX}}}$ in (3.21), and when t_{go} is very small, $\Phi_i \geq M \approx 0$ is trivially satisfied. Therefore, the proposed sliding parameter satisfies the conditions for PFTS set by (3.29) and (3.30), with settling time bounded by (3.28). Thus, even when the global finite time stability of Theorem 3.4 is not satisfied, the trajectories of the lander are PFTS, with settling time bounded by (3.28). ■

3.4 SIMULATIONS AND DISCUSSIONS

3.4.1 Simulation Setup and Parameters

To demonstrate the effectiveness of MSS-OTALG, results from simulations are presented in this section. A point-mass lander with specific impulse I_{sp} = 225 s, T_{max} = 31000 N is assumed. The value of T_{max} has been chosen based on the necessary condition derived in Section 2.3.3. The desired terminal states are $\mathbf{r}_f^d$ = [0, 0, 0]^T m, $\mathbf{v}_f^d$ = [0, 0, 0]^T m/s, at terminal time t_f = 100 s. The simulation is stopped when r_z = 0.05 m or desired terminal time is achieved.

To emulate a trench surrounding the landing site on Mars, the terrain is modelled as a 2-step, flat-top shape (similar to the illustration in Fig. 3.2). The height and width of

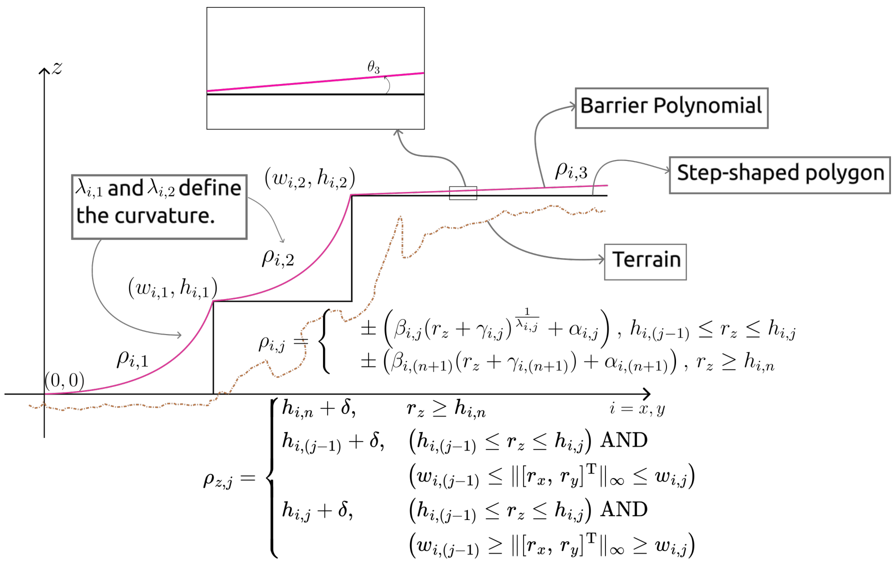
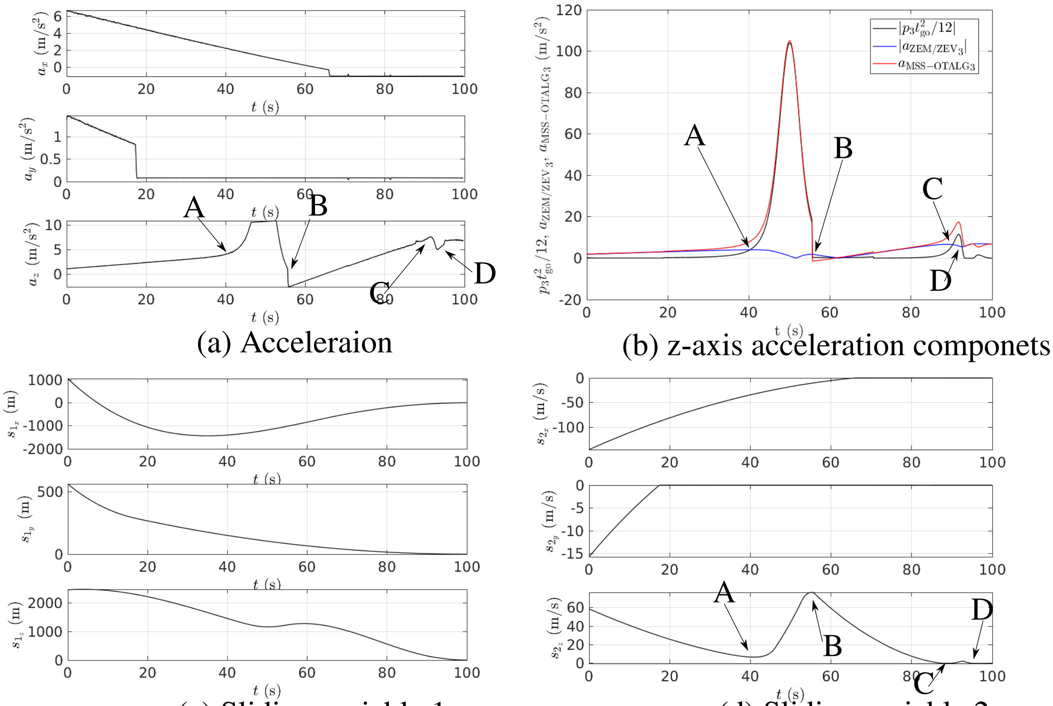


Figure 3.2: Illustration of barriers with $n = 2$.

each step from the origin are given as $h_{i,i} = 500,\ h_{i,2} = 1000,\ w_{i,1} = 600,\ w_{i,2} = 1000$m. To design the barriers, $\theta_3 = 0.05°$, with $\lambda_{i,2} = 6$, and $\lambda_{i,1} = 20$ are chosen. The guidance law constant are chosen as $l_{1,i} = 1,\ l_{2,i} = 9500,$ and $l_{3,i} = 500$, which gives the margin of safety for the vertical motion barrier as $\delta = 1.2 \cdot d_i^* = 95.5$m. The local gravity at Mars is assumed to be $\mathbf{g} = [0,\ 0,\ -3.7114]^{\mathrm{T}}$ m/s^2, and acceleration due to gravity on Earth $g_e = 9.807$ m/s^2 [1]. Thruster actuation latency is incorporated as first-order delays as $\dot{\mathbf{a}}_c = (\mathbf{a}_{c_{\text{ideal}}} - \mathbf{a}_c)/\tau$ [38] where $\tau = 0.0556$s to emulate 90% step response in 50ms [9]. In reality, the thrust commanded in never the exact thrust generated, especially in the case of solid rocket motors. To emulate this, the thrust command is perturbed by ±5% using MATLAB's `rand()` command. Finally, for sliding mode control, $\Phi_i = k_1|p_i|t_{\text{go}}^2/12 + k_2 a_{p_{\text{MAX}}}$ where $k_1 = 0.8$ and $k_2 = 0.2$. To avoid the chattering problem associated with the signum function, the saturation function with boundary layer width $\epsilon = 0.1$ is used.

3.4.2 Illustration of a Numerical Example

To showcase the nominal performance of the proposed guidance law, the simulation results are presented in Figure 3.3. The initial conditions for this simulation are:



(a) Acceleraion (b) z-axis acceleration componets

(c) Sliding variable 1 (d) Sliding variable 2

Figure 3.3: Trajectory, velocity, commanded acceleration, divert term and sliding variables. [A: Divert manoeuvre 1 begins, B: Divert manoeuvre 1 ends, C: Divert manoeuvre 2 begins, D: Divert manoeuvre 2 ends]

$\mathbf{r}_0 = [1051.86,\ 562.15,\ 2459.07]^{\mathrm{T}}$ m, $\mathbf{v}_0 = [-165,\ -26.91,\ 9.45]^{\mathrm{T}}$ m/s and $m_0 = 1905$ kg. From the trajectories, position, velocity and the commanded acceleration plots in Figs. 3.3a-d, it may be observed that the lander rises initially with a positive v_z and then begins to descend at $t = 4$s under the influence of gravity. During this time, to slow down the descent, a positive a_z is continued to be commanded by the guidance law. Besides, note that a large and negative v_x causes the lander to overshoot the desired landing site along the x-direction, which prompts the guidance law to generate a large and positive a_x in order to slow down the lateral motion and bring the lander towards the landing site. As the lander nears the vertical motion barrier at $z = 1000$m (refer to Figs. 3.3a-b), the first dominant divert manoeuvre begins at $t = 40.2$s as the divert term surpasses the ZEM/ZEV term in the overall acceleration command (refer to (2.24), (3.11)), which is evident from Fig. 3.3e. As the vertical motion barrier is encountered, the divert acceleration and hence the commanded acceleration ramps up smoothly and does not exhibit discontinuities in the a_z profile, which is justified from (2.15) and Fig. 3.1. Meanwhile, under the influence of positive a_x, the lander crosses the $x = -1000$m mark at $t = 55.5$s. At this point, the vertical motion barrier switches from $\rho_{z,3} = 1000 + \delta$ to $\rho_{z,2} = 500 + \delta$. As a consequence, the magnitude of divert term falls below the magnitude of ZEM/ZEV term, and the first dominant divert manoeuvre comes to an end (refer to Fig. 3.3e). This also leads to the discontinuity observed in the a_z profile at $t = 55.5$s. On the other hand, the small discontinuities observed in the a_x and a_y profiles (refer to Fig. 3.3d) are due to s_{2x} and s_{2y} reaching nearly zero at around $t = 65.8$s and $t = 17.3$s, respectively, as can be seen in Fig. 3.3g. Those time-instants onwards, very small magnitude of a_x and a_y are only commanded to maintain s_{2x} and s_{2y}, respectively, close to zero.

The effect of the divert term can also be observed in the sliding variables s_{1z} and s_{2z}. Recall from (3.21) that the rate at which the s_{2z} converges to zero is dependent on the choice of k_1 and k_2 in the sliding parameter. Larger values of of k_1 and k_2 imply that the convergence of s_{2z} to zero is more aggressive. However, to execute the divert manoeuvre whenever necessary, smaller value of k_1 and k_2 are desirable in order to allow the state-space trajectory to leave the neighbourhood of the sliding surface $s_{2z} = 0$. Hence, $k_1 = 0.8,\ k_2 = 0.2$ are chosen in this simulation.

Similar to the first divert manoeuvre, as observed from Fig. 3.3e, the second divert manoeuvre takes place when the lander approaches the last vertical motion barrier $\rho_{z,1} = \delta$ near the landing site. This slows down the lander further, thus further facilitating soft landing. However, the divert manoeuvre, in this case, ends soon due to small t_{go}. However, when the lander crosses this barrier, the sign of divert term changes, as expected from its expression in (2.15). To avoid this behaviour, it is recommended that $\rho_{z,1}$ be sufficiently close to r^d_{fz}.

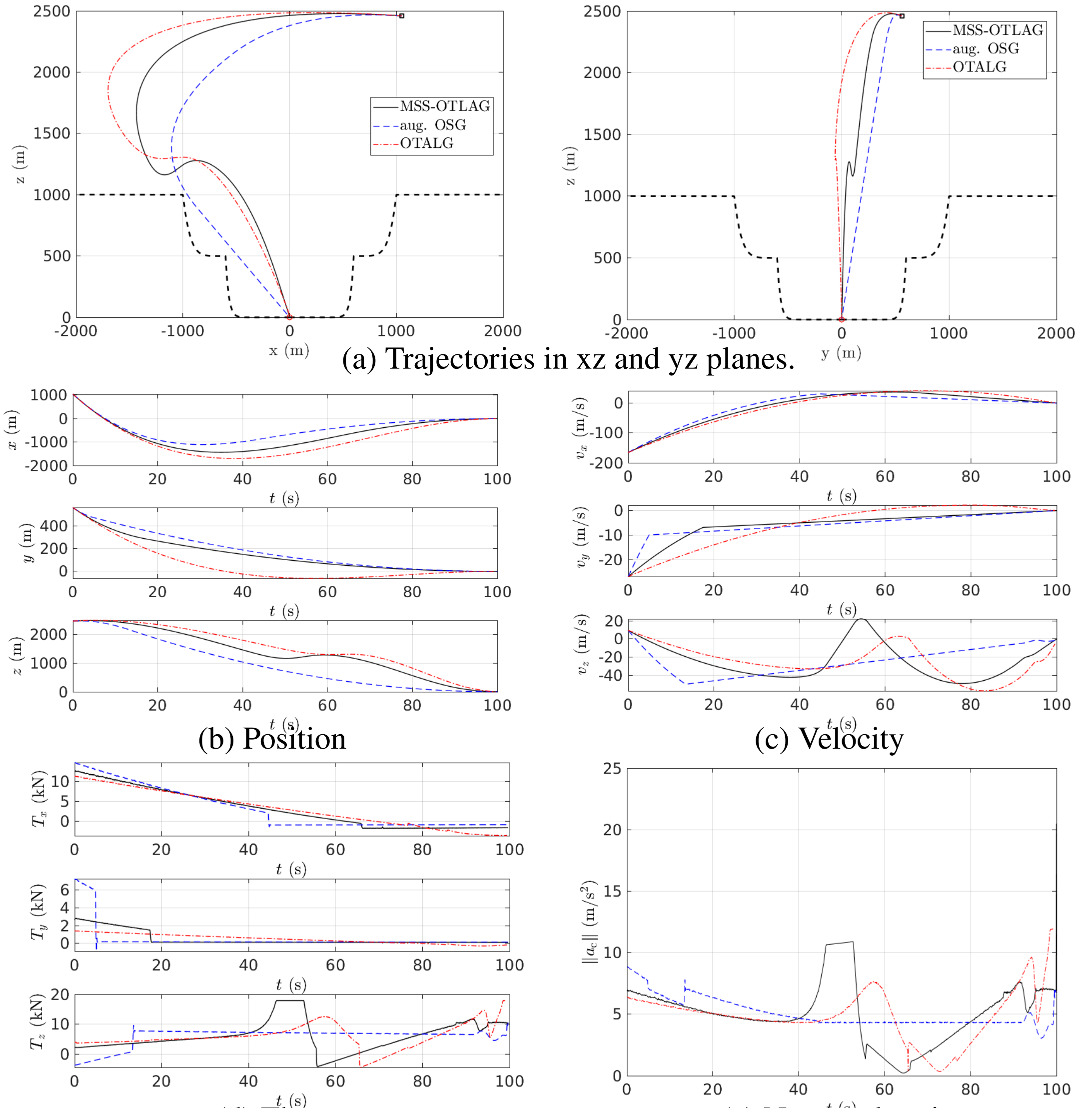
In this numerical example, there are two dominant divert manoeuvres. Since the number of divert manoeuvres is finite, and the sliding variables reach zero in finite time (see Figs. 3.3f-g), robustness of the MSS-OTALG is validated following the notion of PFTS in Theorem 3.5.

3.4.3 Comparative Simulation Study

Very few papers in existing literature have addressed both the problems of precision soft landing and terrain avoidance simultaneously. In this regard, it may be noted that it was stated in [38], a widely-referred precision soft landing paper, that the OSG presented therein could be augmented with the method in [45] for achieving precision soft-landing with terrain avoidance. This augmentation methods was later improved in [43]. Also, the MSS-OTALG presented in this paper is an expansion over the OTALG of previous chapter, which also dealt with both these problems in an integrated way. Thus, in this section, the simulation study in Section 3.4.2 is extended to incorporate a comparative analysis of the performance of the MSS-OTALG w.r.t. that of the OTALG and augmented OSG [38; 43]. Illustrative examples of this comparative study using the same initial conditions and desired terminal condition, as in the previous subsection, under zero and non-zero atmospheric perturbation are presented in Figs. 3.4 (in Section 3.4.3) and 3.6 (in Section 3.4.3), respectively. Subsequently, extensive comparison study results under both zero and non-zero atmospheric perturbation are presented in Figs. 3.5 and 3.7, respectively.

Comparison study under zero atmospheric perturbations

From the trajectories in Fig. 3.4a, position profiles in Fig. 3.4b and velocity profiles in Fig. 3.4c, it is observed that all three guidance laws under comparison are able to drive the lander towards the desired landing site precisely and softly, while also successfully avoiding the terrain. From the net acceleration profile in Fig. 3.4e, observe that the augmented OSG applies a larger initial acceleration to bring the trajectory close to the sliding surface till $t = 44.6$s and subsequently a nearly constant acceleration to maintain the trajectory near the sliding surface. However, as the augmented OSG has been formulated to avoid terrain only in the $z-$direction, it avoids the terrain only marginally in the $x - y$ direction, as can be observed in Fig. 3.4a. On the other hand, the OTALG is formulated to avoid the terrain in any direction. When the lander is away from any terrain, OTALG behaves similar to the OGL [10] in the sense that it applies just enough acceleration for precision soft-landing. But, when the terrain is encountered, the divert term starts to dominate the ZEM/ZEV term in the guidance law, leading to a higher acceleration commanded by OTALG to avoid the terrain and again bring it back to the desired landing site when the terrain is sufficiently avoided. The consequence of these two very different acceleration profiles is that while the augmented OSG has an excellent performance in terms of landing precision but the OTALG outperforms the former in terms of fuel consumption and terrain avoidance. Further, in the case of augmented OSG, increasing the sliding parameter to improve precision also increases the turn rate of the lander, which may be detrimental to the sensitive equipment carried onboard. To this end, the MSS-OTALG developed in this paper finds a middle ground between these conflicting objectives.



(a) Trajectories in xz and yz planes.

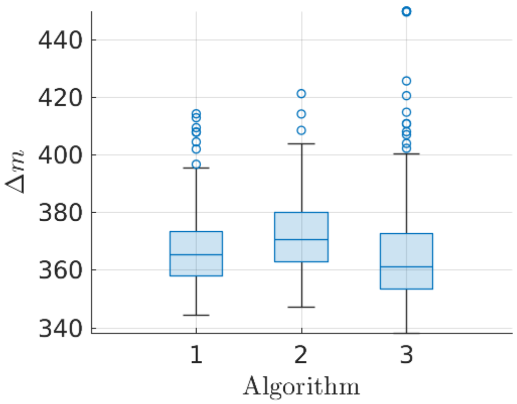
(b) Position

(c) Velocity

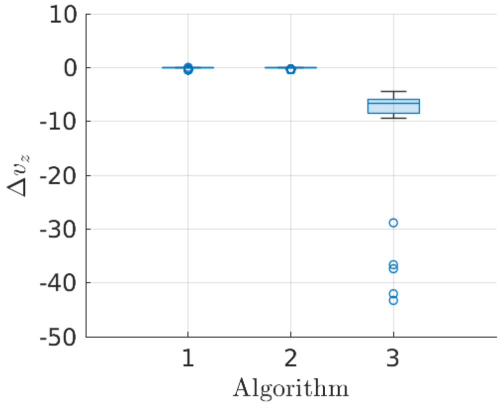
(d) Thrust

(e) Net acceleration

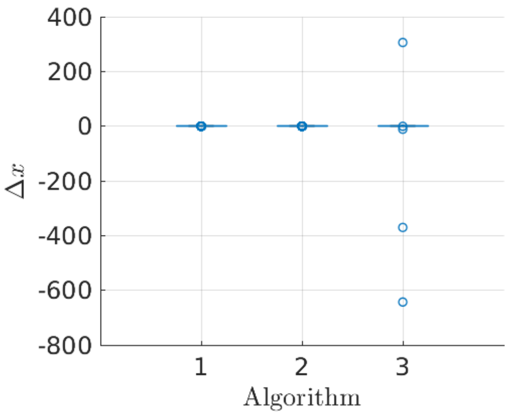
Figure 3.4: Comparison of MSS-OTALG, aug. OSG and OTALG under no atmospheric perturbations.

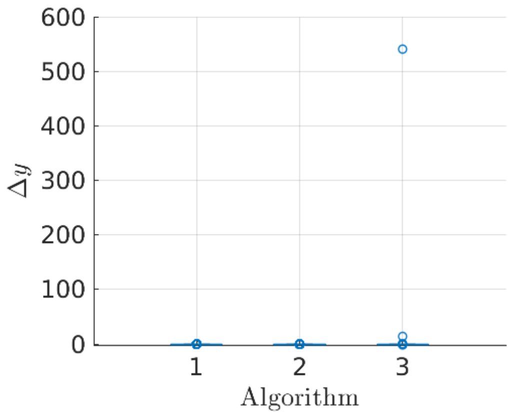


(a) Fuel Consumption



(d) Terminal Descent Velocity





(b), (c) Landing Precision in $x-$ and $y-$ direction

Figure 3.5: MC simulation results for fuel consumption, landing dispersion and residual terminal velocity ($a_P = 0$). [1: MSS-OTALG, 2: augmented OSG, 3: OTALG]

Table 3.1: Normal Distribution for Initial Conditions

State	x_0	y_0	z_0	v_{x0}	v_{y0}	v_{z0}	m_0
Mean	0	0	2500	0	0	-80	1905
SD	2200	2200	400	80	80	20	0

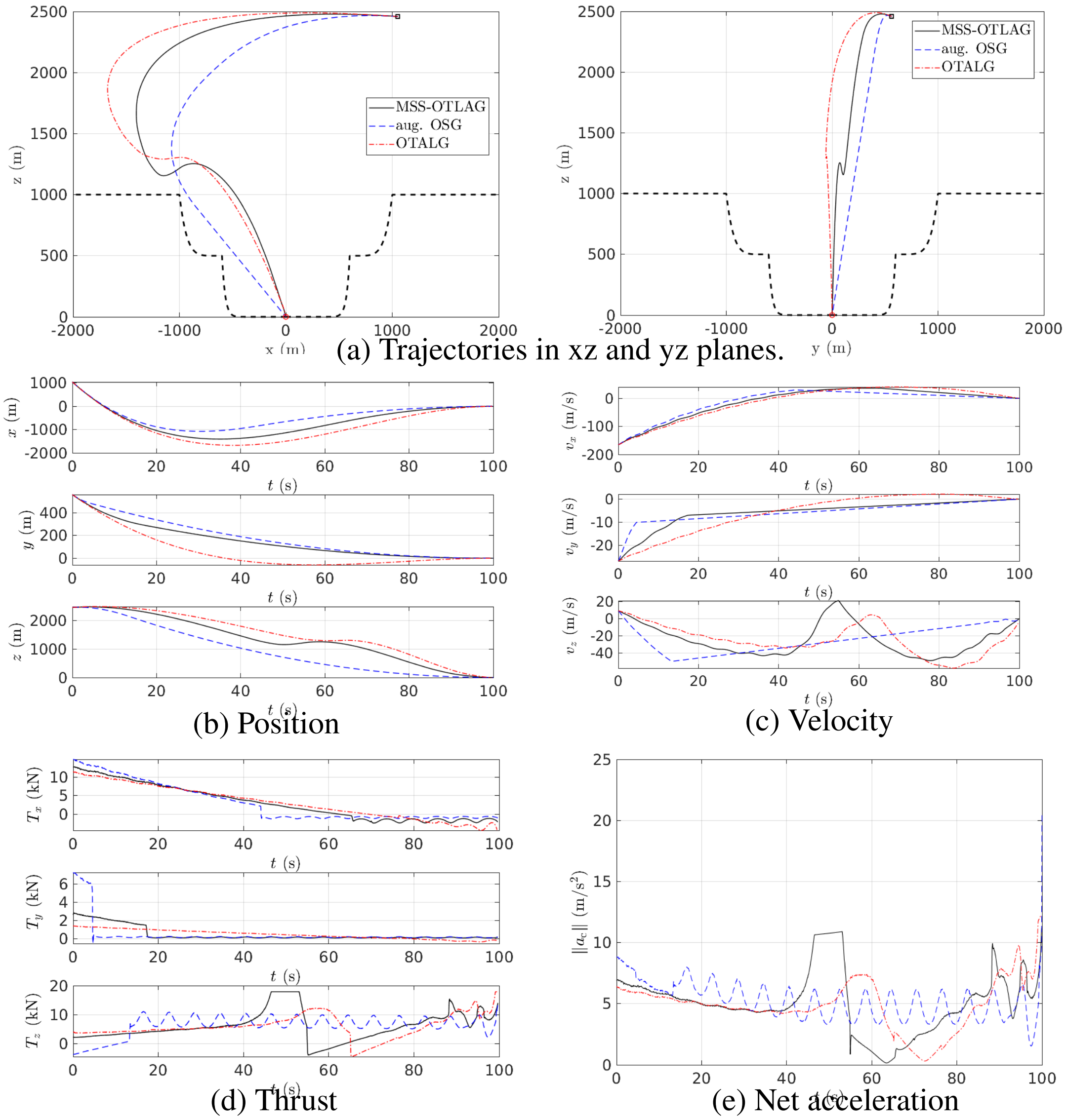
The presence of the sliding term in the MSS-OTALG imparts a high degree of precision, akin to the augmented OSG to the tune of $\Delta x_f = 9.31 \cdot 10^{-7}$m, $\Delta y_f = 3.64 \cdot 10^{-7}$m and $\Delta v_{zf} = -0.02$m/s for MSS-OTALG, $\Delta x_f = -9.98 \cdot 10^{-6}$m, $\Delta y_f = 3.45 \cdot 10^{-6}$m and $\Delta v_{zf} = -0.01$m/s for augmented OSG and $\Delta x_f = -7.32 \cdot 10^{-2}$m, $\Delta y_f = 3.0 \cdot 10^{-3}$m and $\Delta v_{zf} = -3.67$m/s for OTALG. Moreover, as the OTALG is also embedded in the formulation of MSS-OTALG, it inherits the feature of terrain avoidance in all directions and near-fuel-optimality from OTALG, which is evident from fuel consumption data ($\Delta m = 391.37$ kg for MSS-OTALG, $\Delta m = 394.60$ kg for aug. OSG and $\Delta m = 379.22$ kg for OTALG). In this way, MSS-OTALG reconciles seemingly conflicting objectives, with both the sliding and divert terms collaborating to achieve terrain-avoided soft landing objectives. Monte Carlo simulations are conducted using 300 initial conditions selected from the normal distribution outlined in Table 3.1. The purpose is to objectively assess the soft-landing accuracy and fuel consumption statistics of the three guidance laws under comparison. The results of these simulations are depicted using box plot representation in Fig. 3.5, while the corresponding statistical data is summarised in Table 3.2. Augmented OSG and MSS-OTALG exhibit a high degree of accuracy in precision soft landing (refer to Fig. 3.5b-d), yet the former consumes significantly more fuel than the latter (as determined via paired t-test with null hypothesis $H_0 : \Delta m_{\text{aug. OSG}} = \Delta m_{\text{MSS-OTLAG}}$ against $H_1 : \Delta m_{\text{aug. OSG}} > \Delta m_{\text{MSS-OTLAG}}$, which gives $t_{\text{stat}} = 14.35$). This can also be observed from Fig. 3.5a. Conversely, OTALG is found to consume quite less fuel compared to MSS-OTALG, but performs poorly in terms of precision soft-landing performance. Thus, these Monte Carlo studies validate that the MSS-OTALG developed in this paper achieves significantly superior performance in precision soft-landing while also avoiding terrain yet demanding near-to-optimal fuel consumption.

Table 3.2: Terminal States Statistics ($a_{\mathrm{p}} = 0$)

Guidance Law	Mean				SD			
	Δm	Δx	Δy	Δv_z	Δm	Δx	Δy	Δv_z
MSS-OTALG	366.67	$1.65 \cdot 10^{-5}$	$4.37 \cdot 10^{-5}$	$-3.32 \cdot 10^{-2}$	12.72	$6.25 \cdot 10^{-4}$	$6.48 \cdot 10^{-4}$	$8.29 \cdot 10^{-2}$
aug. OSG	371.73	$1.67 \cdot 10^{-6}$	$-1.55 \cdot 10^{-5}$	$-6.69 \cdot 10^{-2}$	12.89	$6.78 \cdot 10^{-4}$	$6.19 \cdot 10^{-4}$	0.13
OTALG	365.91	−2.40	1.84	−7.43	19.62	46.29	31.23	4.22

Comparison study under non-zero atmospheric perturbations

Presence of disturbances, such as those caused due to atmosphere, can cause the lander to go off the nominal course and can cause the lander to perform poorly in terms of precision soft landing. In this section, comparative simulation study under non-zero atmospheric perturbations is presented in which $\mathbf{a}_p(t) = 0.3\mathbf{a}_c \sin(\pi t/3)$ is considered as the model for atmospheric perturbations. Using the same initial conditions as in Fig. 3.4, illustrative examples using the three guidance laws - MSS-OTALG, augmented OSG and OTALG - are presented in Fig. 3.6. All of them are able to avoid the terrain and precisely and softly land close to the desired landing site, as evident from Fig. 3.6a-c. Similar to the zero perturbation case, the augmented OSG demands a higher initial acceleration due to the sliding term, which in effect improves the disturbance rejection capability as can be observed in Fig. 3.6d-e, where the command acceleration and thrust vary to attenuate the disturbances caused by the winds. This results in almost no oscillations in the velocity of the lander, as can be observed in Fig. 3.6c. On the other hand, since the OTALG does not have any disturbance rejection capability, the thrust and commanded acceleration profiles are similar to that of zero perturbation case, however this causes the lander to sway continuously and thus degrade the soft-landing accuracy, as can be observed in the velocity profiles shown in Fig. 3.6c. Coming to the MSS-OTALG, sufficient disturbance rejection can be observed due to the sliding term, and at the same time the terrain is avoided due to the divert term. Since the constants $k_1,\ k_2 < 1$ have been chosen for the sliding parameter, the disturbance rejection by MSS-OTALG is not



(a) Trajectories in xz and yz planes.

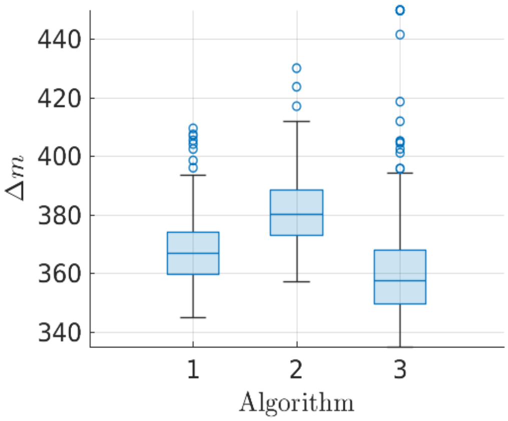
(b) Position

(c) Velocity

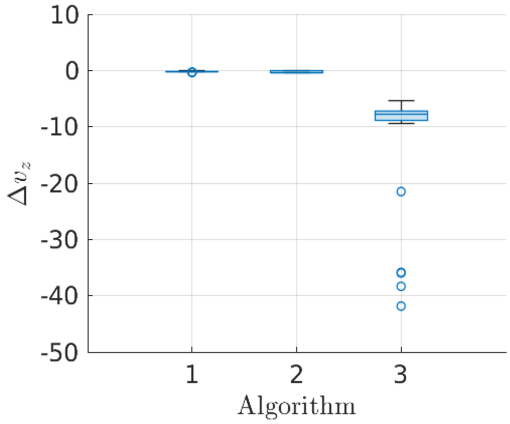
(d) Thrust

(e) Net acceleration

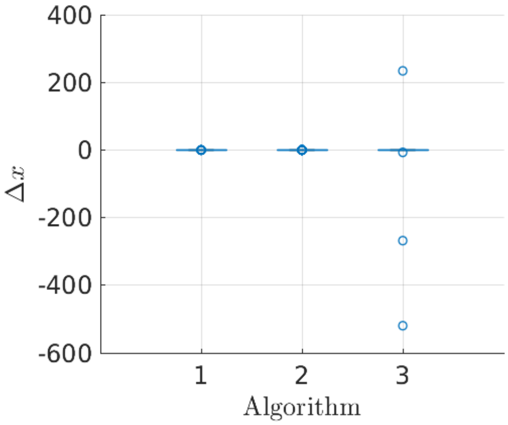
Figure 3.6: Comparison of MSS-OTALG, aug. OSG and OTALG under bounded atmospheric perturbations.

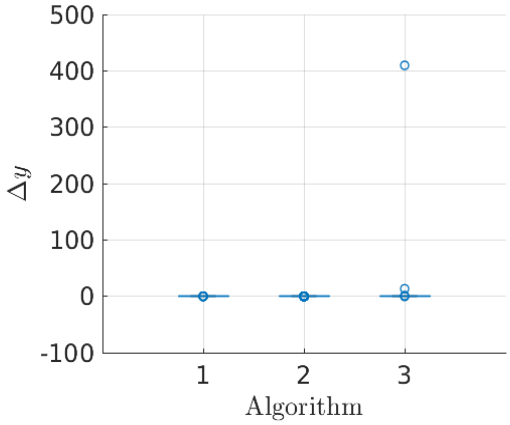


(a) Fuel Consumption



(d) Terminal Descent Velocity





(b), (c) Landing Precision in $x-$ and $y-$ direction

Figure 3.7: MC simulation results for fuel consumption, landing dispersion and residual terminal velocity ($a_P \neq 0$). [1: MSS-OTALG, 2: augmented OSG, 3: OTALG]

as effective as that by the augmented OSG. However, it attenuates the disturbances caused by the wind significantly better that the OTALG. This phenomenon can be observed in the acceleration command and thrust profiles (refer to Fig. 3.6d-e). It can also be observed that as the divert term's influence increases, it dominates the effect of the sliding term to avoid the terrain. Then, when the divert term is small the sliding term operates to mitigate disturbances, improving the accuracy of the precision soft landing.

To assess both the soft-landing precision and fuel usage in the presence of atmospheric disturbances, Monte-Carlo simulations has been conducted utilising the same 300 initial conditions as used in the Monte-Carlo simulations shown in Fig. 3.5. The outcomes are graphically represented via box plots in Fig. 3.7 and numerically summarised in Table 3.3. It's evident from the simulations that the augmented OSG, as anticipated from the thrust and acceleration profiles in Fig. 3.6d-e, consumes more fuel. However, it consistently delivers the most accurate precision soft-landings. MSS-OTALG exhibits slightly higher fuel consumption compared to OTALG but, showcases commendable performance in precision soft-landing akin to the augmented OSG. On the other hand, although OTALG demonstrates lower fuel consumption compared to the augmented OSG and MSS-OTALG, it suffers from significantly poorer performance in precision soft-landing. Hence, the Monte-Carlo simulations effectively validate the robustness of MSS-OTALG in mitigating the impact of non-zero atmospheric perturbations while achieving the main objectives of terrain-avoided precision soft landing.

Table 3.3: Terminal States Statistics ($a_\mathrm{p} \neq 0$)

Guidance Law	Mean				SD			
	Δm	Δx	Δy	Δv_z	Δm	Δx	Δy	Δv_z
MSS-OTALG	367.73	$2.75 \cdot 10^{-5}$	$-4.71 \cdot 10^{-5}$	-0.17	11.74	$1.37 \cdot 10^{-3}$	$1.37 \cdot 10^{-4}$	$4.49 \cdot 10^{-2}$
aug. OSG	381.33	$3.03 \cdot 10^{-5}$	$-9.02 \cdot 10^{-5}$	-0.19	12.46	$1.05 \cdot 10^{-3}$	$1.02 \cdot 10^{-3}$	0.15
OTALG	361.50	-1.87	1.41	-8.21	19.58	36.36	23.66	3.71

3.5 CONCLUSION

Expanding upon the Optimal Terrain Avoidance Landing Guidance Law (OTALG) recently presented in literature with the help of sliding mode control by multiple sliding surfaces (MSS), a near-fuel optimal and robust guidance law for precision soft-landing in hazardous terrain, named MSS-OTALG, is presented in this paper. The incorporation of MSS renders it the desired robustness. To allow the lander to manoeuvre away from the terrain, a state and time-dependent sliding parameter is introduced, and practical fixed time stability is proven under the proposed guidance law. Finally, extensive computer simulations validate the ability of the MSS-OTALG to avoid the terrain and precisely and softly land at the desired landing site while having low fuel consumption, under realistic limitations posed by thruster dynamics, thrust constraints and atmospheric disturbances. When compared against the OTALG and an augmented version of optimal sliding guidance (OSG) using Monte Carlo simulations, it was observed that the MSS-OTALG finds the middle ground in terms of all the performance measures.

CHAPTER 4

ONLINE TERRAIN BARRIER FUNCTIONS FOR MSS-OTALG

The results presented in previous chapters require that the barrier functions be pre-computed. And while these barriers can be designed using orbital images, they are only an estimate of the terrain. When embarking on a mission to an area with low-quality orbital images and significant uncertainty about the terrain at the landing site, relying solely on these will not be sufficient. Firstly, factors like imaging artefacts and shadows can distort terrain approximations, diminishing their accuracy. Secondly, creating barrier functions based on these inaccuracies could result in overly cautious or even hazardous outcomes. For such missions, a online navigation and guidance allows us to expand the free region of safe motion for the spacecraft.

For such missions, an integrated terrain avoidance and guidance law is needed, which works using the onboard sensed data. In this chapter, assuming that the elevation map, registered to the world coordinates, from a downward-facing camera is already available,"Online Terrain Barrier Update" (OTBU) algorithm is presented to generate the terrain barrier functions required by MSS-OTALG, on the fly. The terrain elevation information, in the form of a grid map, is fed into OTBU along the estimated highest peak in the neighbourhood of the landing site to generate terrain barriers as polynomial functions. These functions are updated at regular intervals, depending on the distance to the desired landing site. The terrain barriers are then fed into the MSS-OTALG law to generate necessary divert commands to avoid the terrain and achieve high accuracy in precision soft landing. For sake of completeness, the MSS-OTALG law is given as:

$$\mathbf{a} = \frac{6}{t_{\mathrm{go}}^2}\mathbf{ZEM} - \frac{2}{t_{\mathrm{go}}}\mathbf{ZEV} + \mathbf{p}\frac{t_{\mathrm{go}}^2}{12} - \mathbf{\Phi}\,\mathrm{sgn}\,\mathbf{s}_2 \tag{4.1}$$

4.1 ONLINE TERRAIN BARRIER UPDATE

To expand the free-space of safe motion for the spacecraft, in this section, an algorithm to update the polynomial barrier functions, based on the local terrain information, is proposed. Online Terrain Barrier Update (OTBU), in Algorithm 1, leverages information not just from the orbital images, but also from the onboard camera to generate barrier functions ρ_i, which can then be used by MSS-OTALG law (4.1) to manoeuvre about the terrain and safely land at the desired landing site. Here, it is assumed that the local terrain observations are made using an ideal, downward pointing camera. The barrier computation frequency is set to be lower than the camera's observation frames per seconds to reduce overall computational burden. However, the barrier update frequency should not be so slow that the terrain features are missed. Further, the barrier update frequency is needed to be increased as the spacecraft approaches the desired landing site for more accurate navigation and guidance.

4.1.1 Development of the algorithm

From the elevation map made using orbital images, an estimate of the highest point below the powered descent initiation height and above the desired landing site is pre-determined and stored *a-priori* as $\mathbf{r}_{\mathrm{H}}$. The proposed algorithm updates the barrier functions with a frequency based on design requirements. The time at which the barriers were last update is denoted as t_{last} in Algorithm 1, and the barrier computation intervals are $0 < \Delta t_1 < \Delta t_2$. When the spacecraft is more that Θ distance away from the desired landing site, the barriers are updated less frequently, that is with a longer interval Δt_2. And when the spacecraft is closer to landing site, a shorter interval Δt_1, is used. This helps in reducing the number of times the Algorithm 1 is invoked and therefore saves computational costs.

The OTBU procedure is initialised with the estimated highest point $\mathbf{r}_{\mathrm{H}}$, desired landing site $\mathbf{r}_f^d$, last update time $t_{\mathrm{last}} = 0$, safety margin δ, barrier update intervals Δt_1 and Δt_2, update interval switching distance Θ, erroneous peak suppression level α and order of the barrier function m. Once initialised, the OTBU algorithm requires the processed

Algorithm 1 Online Terrain Barrier Update

Require: $\mathbf{r}_{\mathrm{H}}$, $\mathbf{r}_f^d$, t_{last}, δ, $0 < \Delta t_1 < \Delta t_2$, Θ, α, m

procedure OTBU($\mathcal{F}$,$\mathbf{r}$,t)
  if $\|\mathbf{r} - \mathbf{r}_f^d\| < \Theta$ **then**
    $\Delta t \leftarrow \Delta t_1$
  else
    $\Delta t \leftarrow \Delta t_2$
  end if
  if $t - t_{\mathrm{last}} \geq \Delta t$ or $t = 0$ **then**
    $t_{\mathrm{last}} \leftarrow t$
    im $\leftarrow$ `ExtendedMaxima`($\mathcal{F}, \alpha$)
    s $\leftarrow$ `GetCentroids`(im)
    $\mathbf{r}_{\mathrm{peak}} \leftarrow \mathtt{max}(s)$
    if $r_z - r_{\mathrm{H}z} > \delta$ **then**
      $\rho_z \leftarrow r_{\mathrm{H}z} + \delta$
      $\mathbf{b} \leftarrow \mathbf{r}_{\mathrm{H}}$
    else
      $\rho_z \leftarrow r_{\mathrm{peak}_z} + \delta$
      $\mathbf{b} \leftarrow \mathbf{r}_{\mathbf{peak}}$
    end if
    for $v = x,\ y$ **do**
      if $b_v > r_{fv}^d$ **then**
        $i_v \leftarrow \frac{(b_v - r_{fv}^d)^m}{b_z - r_{fz}^d}$
        $j_v \leftarrow (b_v - r_{fv}^d)^m - i_v b_z$
        $k_v \leftarrow r_{fv}^d$
        $\rho_v \leftarrow (i_v r_z + j_v)^{\frac{1}{m}} + k_v$
      else if $b_v < r_{fv}^d$ **then**
        $p_v \leftarrow \frac{(-b_v + r_{fv}^d)^m}{b_z - r_{fz}^d}$
        $q_v \leftarrow (-b_v + r_{fv}^d)^m - p_v b_z$
        $t_v \leftarrow -r_{fv}^d$
        $\rho_v \leftarrow -((p_v r_z + q_v)^{\frac{1}{m}} + t_v)$
      else
        $\rho_v \leftarrow$ `Inf`
      end if
    end for
  end if
end procedure

terrain information $\mathcal{F}$ determined by the onboard camera, current position in inertial coordinates $\mathbf{r}$ and current time t to update the barrier functions. The terrain information $\mathcal{F}$ is assumed to be a grid map with each pixel information registered to its corresponding coordinates in the inertial frame.

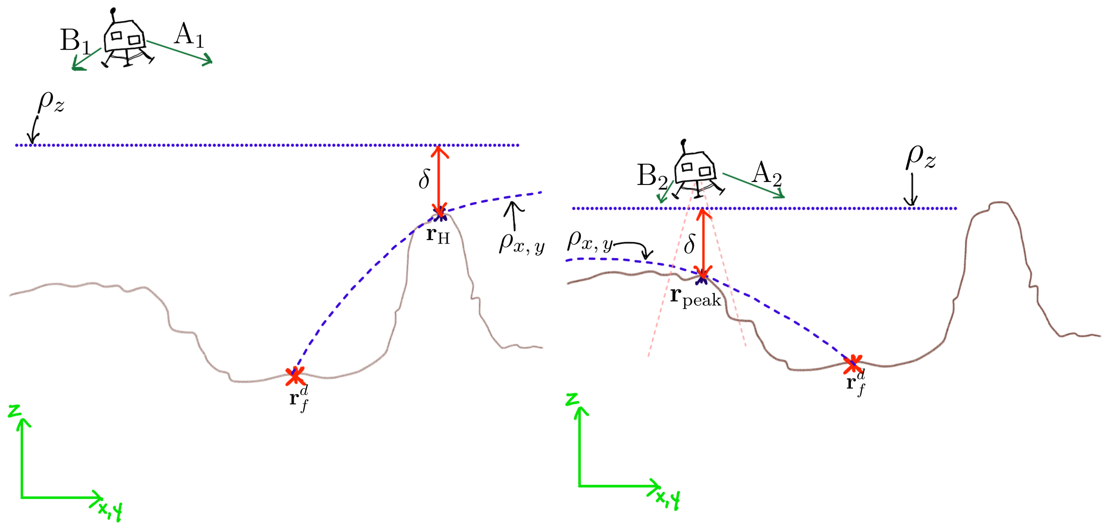
In Lines 2-6 of Algorithm 1, the barrier computation frequency is set depending on how far the lander is from the desired landing site, that is if the spacecraft is farther than Θ distance from desired landing site, the longer update interval Δt_2 is used and Δt_1 is used otherwise. The barriers are updated only if this is the first time barrier functions are being generated, or if the barrier update interval has elapsed since the last computation of barrier functions. The goal of Lines 9-11 is to determine the coordinates of the highest point in $\mathcal{F}$, that is the local terrain information. This is done by first finding peaks in the terrain visible in the onboard camera field-of-view (FOV). In general, the terrain information captured in $\mathcal{F}$ is noisy and have many artefacts. To ignore the artefacts and extract only the useful maxima, H-maxima transformation [36] is done and its regional maxima is taken. The H-maxima transform filters out useful local maxima from the noise by suppressing peaks that have intensity less than a certain value, and then finds the maxima in the remaining regions. In OTBU, this is done in line 9 by the function `ExtendedMaxima`$(\mathcal{F}, \alpha)$ where α is the threshold for suppressing erroneous peaks. In our simulations, this function is implemented using `imextendedmax()` in MATLAB [28]. Then, the centroids of these local maxima are determined in line 10, and the coordinates of the highest centroid is denoted as $\mathbf{r}_{\text{peak}}$.

In lines 12-18, the barrier for vertical motion is defined. Algorithm 1, initially, uses the altitude of the pre-estimated highest point and sets $\rho_z = r_{\text{H}z} + \delta$ as the vertical motion barrier where δ is the safety margin δ. This barrier is maintained till the altitude of the spacecraft is above $r_{\text{H}z} + \delta$. Once this altitude is crossed, the algorithm the altitude $r_{\text{peak}z}$ is used and $r_{\text{peak}z} + \delta$ is set as the vertical motion barrier. This is done to reduce the number of changes in the barriers experienced by the MSS-OTALG, and therefore to reduce the excessive switching in guidance command.

Along the similar lines, a parameter for generating horizontal motion barriers, $\mathbf{b}$, is set in Lines 14 and 17. The parameter $\mathbf{b}$ is then used in Lines 19-29 to generate the barrier functions to arrest the horizontal motion. First, the position of the parameter $\mathbf{b}$ is compared component-wise with the desired landing site. Then, depending on this $\rho_{x,y}$ is set as per Lines 20-23 and 25-28. The barrier functions are designed such that $\rho_{x,y}$ contains both, the desired landing site and the parameter $\mathbf{b}$. This along with suitably chosen value of the constant m, which specifies the order of the barrier functions, are used to shape the barrier functions as dictated by mission requirement, such as conservativeness, safety requirements etc. Further, as the value of m approaches unity, the barriers take the shape of glideslopes, and as m approaches zero, it takes the shape similar to that of the vertical motion barrier, with $m = 0$ being undefined. To ensure the terrain is sufficiently bounded and the barriers are not overly conservative, it suggested that $m \approx 0.5$ be chosen. This approach guarantees the presence of a barrier to halt horizontal movement, regardless of the spacecraft's direction or the terrain's location. Additionally, it ensures that the terrain is adequately avoided without the barriers being overly cautious. After updating all the barrier functions, the algorithm pauses until the update interval has passed, then the flight computer re-invokes the algorithm.

Remark 2. In Algorithm 1, two corner cases have been identified. The first, and the rarer of two, occurs when the spacecraft passes through a canyon-like terrain in its approach towards the desired landing site. In this situation, $b_v = r^d_{fv}$, for $v = x,\ y$. This leads to one of the terrain barriers (ρ_x or ρ_y) to not be defined, depending on weather $b_x = r^d_{fx}$ or $b_y = r^d_{fy}$. However, the other defined horizontal motion barrier and ρ_z remain defined and are able to steer the spacecraft towards the landing site and away from the hazardous terrain.

The next, and more common, corner case is $\mathbf{b} = \mathbf{r}^d_f$ leading to both the horizontal motion barriers to be undefined. However, this situation arises only when the spacecraft is right



(a) Illustration of barrier functions when $r_z - r_{\mathrm{H}z} > \delta$

(b) Illustration of barrier functions when $r_z - r_{\mathrm{H}z} < \delta$

Figure 4.1: Illustration of barrier functions.

above the desired landing site. Since the desired landing site is generally chosen to be sufficiently safe from any hazardous terrain in $x-$ and $y-$ direction, just the vertical motion barriers are sufficient to avoid crashing down on the landing site.

To alleviate these corner cases of undefined horizontal motion barriers by setting the barrier function with a large number causing the corresponding divert acceleration to be small and not effect the guidance law. The vertical motion barrier is not affected by these corner cases and is set as described above.

4.1.2 Discussion on the algorithm

In this section a discussion is presented to look into how the proposed algorithm is able to ensure that the spacecraft is steered away from potential terrain and safely diverted towards the desired landing site for a soft touchdown. Before that, note that the guidance law in Eq. (4.1) has three distinct terms: the landing guidance term (i.e. ZEM/ZEV term), the divert manoeuvre term (i.e. $\mathbf{p}\frac{t_{go}^2}{12}$ term) and the sliding mode term (i.e. $-\boldsymbol{\Phi}\,\mathrm{sgn}\,\mathbf{s}_2$ term). The landing guidance term is responsible for fuel-optimal steering the spacecraft towards the desired landing site. The divert manoeuvre term provides the necessary divert guidance commands to avoid the terrain. And, the sliding mode term imparts robustness to the landing guidance and improves the accuracy of the precision soft landing.

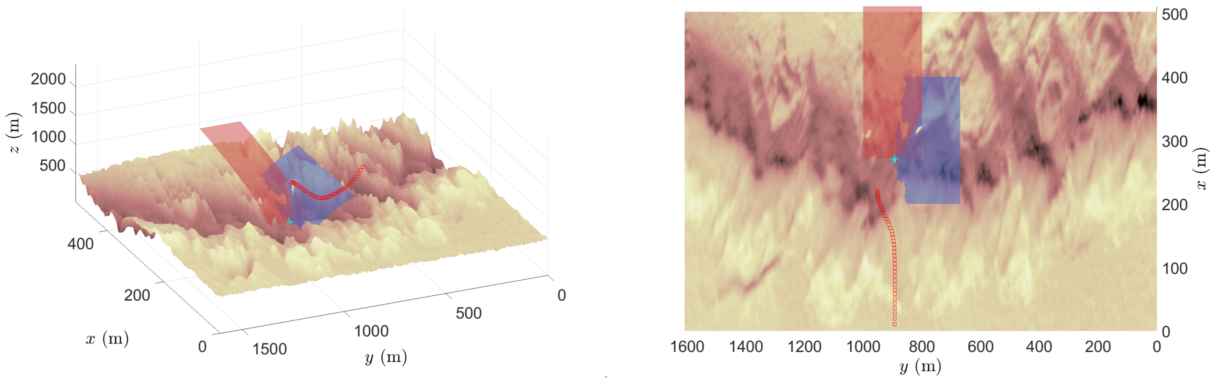
Consider the case in Fig. 4.1a where the altitude of the spacecraft is above $r_{\mathrm{H}z} + \delta$. Suppose the spacecraft is located at the location as shown in the figure, with heading direction B1. In this case, the most imminent terrain barrier is ρ_z. Here, the landing guidance tries to change the direction of horizontal velocity towards $\mathbf{r}_f^d$ and slow down the descent to allow the spacecraft to land softly. Also, due to the presence of ρ_z, the divert term further slows down the descent and ensures that the spacecraft does not crash in to the terrain. As the spacecraft descends and falls below the $r_{\mathrm{H}z} + \delta$ altitude, as shown in Fig. 4.1b. In this scenario, consider the spacecraft is heading along B2. Based on camera observations, the OTBU algorithm described above sets the point $\mathbf{r}_{\mathrm{pk}}$ as the parameter $\mathbf{b}$ and generates the horizontal barrier functions and sets the vertical motion barrier. Since the spacecraft is still moving towards left, it approaches the horizontal barrier which causes the divert term to increase and steer the spacecraft away from the terrain and push it towards the landing site.

Now consider the other scenario in Fig. 4.1a where the spacecraft altitude is above $r_{\mathrm{H}z} + \delta$ but the heading is along A1. In this case, as the spacecraft approaches the hill-like terrain feature from the left. The landing guidance term attempts to steer the spacecraft towards the desired landing site. However, due to the initial heading in this case, it also approaches the horizontal and vertical barrier (as shown in Fig. 4.1a). As a consequence, the divert term suitably generates guidance commands and steers the spacecraft away from the hazardous terrain. In this situation, the motion barriers $\rho_{x,,y}$ and ρ_z not only prevent the spacecraft from crashing into the terrain but also blockade it from reaching a potentially irrecoverable region behind this hill-like terrain. As the spacecraft descends and falls below the $r_{\mathrm{H}z} + \delta$ altitude, the local terrain observations are used to generate the terrain barriers and manoeuvre the spacecraft towards the landing site. Suppose, the spacecraft crosses the $r_{\mathrm{H}z} + \delta$ altitude and flies along the heading A2 as shown in Fig. 4.1b. The spacecraft is heading towards the desired landing site and the OTBU algorithm generates the terrain barrier functions based on the current observations, and the MSS-OTALG law uses it to ensure that a crash landing does not take place. Now, primarily due to the effect of landing guidance term, along with the sliding term, the spacecraft is steered accurately and softly to the desired landing site.

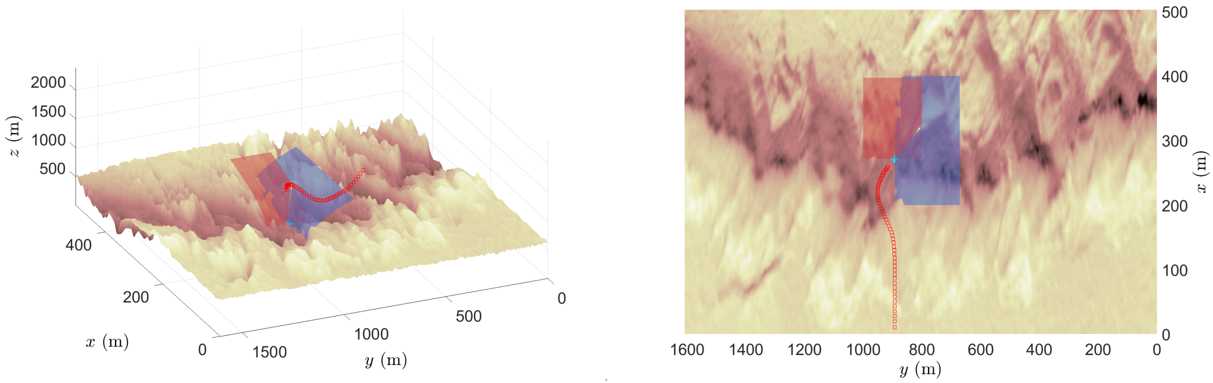
4.2 SIMULATIONS

4.2.1 Simulation Setup and Parameters

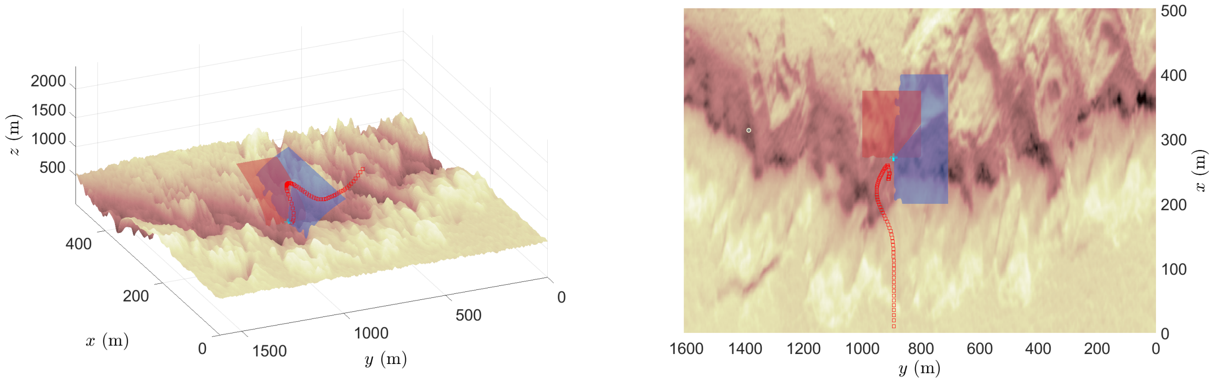
The simulations are done in MATLAB 2023b using Intel Core i5-10500 CPU @ 3.10 GHz with 16 GB of DDR4 memory. The simulations are performed with MSS-OTALG guiding the spacecraft and in this section, the simulation parameters are taken from [4]. Simulation setup parameters have been summarised in Table 4.1. Unlike in [3; 4], the final time is not same for all initial conditions and are chosen based on the initial and final conditions. In this letter, the method proposed in [19] is used to calculate the optimal time-to-go for constrained terminal velocity guidance law, which gives the landing guidance term in (4.1). However, the terminal time calculated using this method does not account for the divert manoeuvres and the time lost due to it. To compensate for this, a buffer time of 20 s is added to the optimal terminal time. Note that the fuel consumption depends heavily on the constants chosen for the sliding parameter and the buffer chosen for the final time. To simulate the terrain in Mars, digital terrain maps (DTM) from the HiRISE database are used. Specifically, the DTM titled "Possible Vent in Ceraunius Fossae" [30] is considered. The DTM is cropped to a 4901x1651 px image, and the data are converted from `uint8` to `double`. The resulting image is then scaled to create an elevation map of maximum altitude of 1000m. This particular DTM has a resolution of 2.01m/px, which is then used to scale the x- and y-coordinates using the image's pixel dimensions.



(a) Barrier functions at $t = 40$s.



(b) Barrier functions at $t = 55$s.



(c) Barrier functions at $t = 65$s.

Figure 4.2: Evolution of barrier functions. Red: ρ_x, Blue: ρ_y.

Table 4.1: Simulation setup parameters

	Parameter	Values
Spacecraft Parameters	Actuator lag time-constant, τ	0.0556 s
	Specific impulse, I_{sp}	225 s
	Max thrust, $T_{\max}$	31000 N
Guidance Parameters	Divert term constants, $l_{1,i},\ l_{2,i},\ l_{3,i}$	1, 9500, 500
	Sliding term constants, $k_1,\ k_2$	0.8, 0.2
	Virtual controller gain, Λ	2
	Safety margin, δ	95.5m
Algorithm Paramters	Update intervals, $\Delta t_1,\ \Delta t_2$	0.05, 0.1 s
	Update interval switching distance, Θ	1000 m
	`ExtendedMaxima` parameter, α	10
	Order of barrier polynomials, m	0.5
	Location of est. highest point, $\mathbf{r}_{\mathrm{H}}$	$[480.39, 751.74, 858.82]^{\mathrm{T}}$

4.2.2 Illustrative Example

To demonstrate the working of MSS-OTALG law in combination with the OTBU algorithm presented in Algorithm 1 (henceforth, termed as 'online MSS-OTALG') and to illustrate the evolution of barrier functions as the spacecraft approaches the desired landing site, simulation results with initial conditions as: $\mathbf{r}_0 = [0, 892.44, 2500]^{\mathrm{T}}$m, $\mathbf{v}_0 = [10, 0, -100]^{\mathrm{T}}$m/s, $m_0 = 1905$kg are presented. The desired landing site is at $\mathbf{r}_f^d = [271.35, 892.44, 280.294]^{\mathrm{T}}$m, and the final time, including the buffer, for this scenario is computed using the method in [19] as $t_f = 69.54$s.

To illustrate the evolution of the terrain barriers obtained by Algorithm 1 and the trajectory governed by MSS-OTALG, three instances of the simulation, at $t = 40,\ 55,$ and 65s, are presented in Fig. 4.2. Each horizontal motion barrier, $\rho_{x,,y}$, is generated in x-z and y-z planes, respectively. These barriers are represented as surfaces in Fig. 4.2 by extending them along the third direction. At the beginning of the descent, when the spacecraft is sufficiently away from the desired landing site (i.e. $\|\mathbf{r} - \mathbf{r}_f^d\| > 1000$m), the online MSS-OTALG uses the longer update interval Δt_2 to update the terrain barriers. Further, since the initial altitude of the spacecraft is more than $r_{\mathrm{H}z} + \delta = 954.32$m, the OTBU algorithm sets $\rho_z = r_{\mathrm{H}z} + \delta$ and the parameter $\mathbf{b} = \mathbf{r}_{\mathrm{H}}$ as described in Lines 13-14,

Table 4.2: Distribution for Initial Conditions

State	x_0	y_0	z_0	v_{x0}	v_{y0}	v_{z0}	m_0
Mean	1650	5000	2500	0	0	-80	1905
SD	1000	1000	400	80	80	20	0

illustrated shown in Fig. 4.2a. These terrain barriers are maintained till $r_z < r_{Hz} + \delta$, occurring at $t = 54.94$s. After this instance, the algorithm switches from using $\mathbf{r}_H$ as the parameter $\mathbf{b}$ to using the $\mathbf{r}_{peak}$ from the local observations from the onboard sensors, as described in Lines 16-17. Now, $||\mathbf{r} - \mathbf{r}_f^d|| < 1000$m. This causes the algorithm to use the shorter update interval Δt_1 to update the terrain barriers. The evolution of these barriers can be observed in the Figs. 4.2b and 4.2c. Observe in Fig. 4.2b, the spacecraft tries to fly in the direction of decreasing y-coordinate, however, due to the presence of ρ_y barrier it is steered away from hazardous terrain and towards the desired landing site. Similarly in Fig. 4.2c, the spacecraft overshoots the landing site in x-direction. This could potentially lead to a crash in the nearby terrain. The ρ_x barrier, however, prevents this, and as the spacecraft approaches the barrier, the divert term in (4.1) of online MSS-OTALG increases and steers the spacecraft away from the terrain. Finally, the ZEM/ZEV term and the sliding mode term in (4.1) facilitate the spacecraft to land softly and precisely.

4.2.3 Statistical studies on performance measures

In this section, the results of Monte Carlo-like simulations are presented. The online MSS-OTALG is used to guide a spacecraft to the landing site located inside a ridge at $\mathbf{r}_f^d = [842.7,\ 3516.2,\ 501.6]^T$m. The 100 initial conditions of the spacecraft are sampled from truncated uniform distribution, where the distribution is truncated at $\mu \pm$ SD. The relevant details of the distribution are given in Table 4.2 and the initial conditions are shown in Fig. 4.3 with desired landing site in cyan. To quantitatively measure the performance of online MSS-OTALG, the following terminal deviations are measured: lateral landing dispersion $\Delta xy = \sqrt{(r_x^d - r_{fx}^d)^2 + (r_y^d - r_{fy}^d)^2}$, terminal velocity deviation

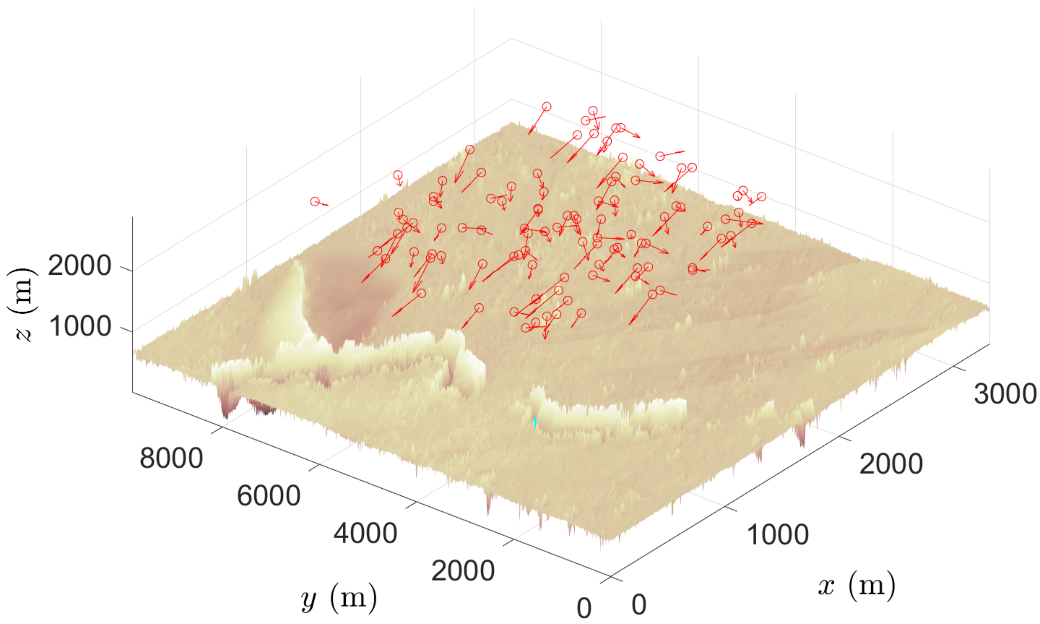
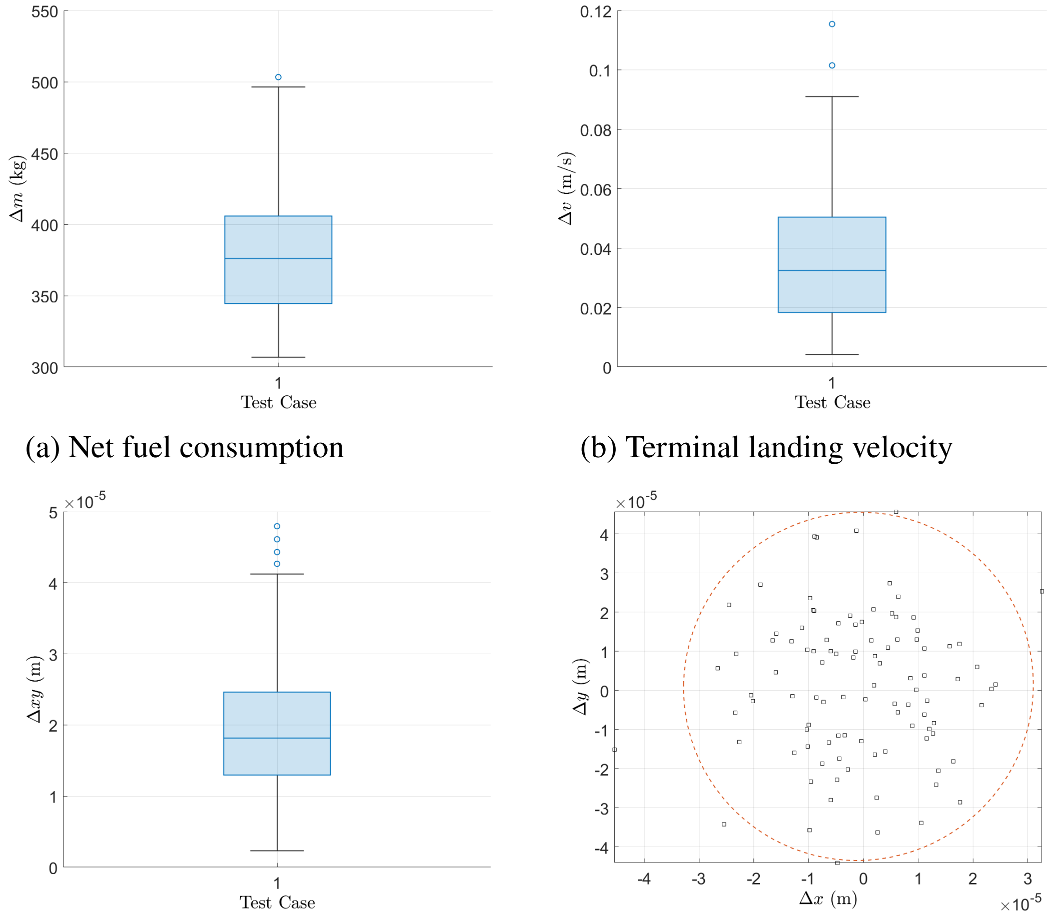


Figure 4.3: Initial conditions for the Monte-Carlo simulations

$\Delta v = \|\mathbf{v}^d - \mathbf{v}_f^d$ and net fuel consumption $\Delta m = m_0 - m_f$. To that end, the terminal statistics are: for net fuel consumption $\mu_{\Delta m} = 380.06$kg and $\sigma_{\Delta m} = 47.79$kg, for terminal landing dispersion $\mu_{\Delta xy} = 2.002 \cdot 10^{-5}$m and $\sigma_{\Delta xy} = 9.91 \cdot 10^{-6}$m, and for terminal landing velocity $\mu_{\Delta v} = 0.036$m/s and $\sigma_{\Delta v} = 0.021$m/s. Both, the qualitative results (as shown in Fig. 4.4) and the aforementioned quantitative results, show that the online MSS-OTALG is able to not only avoid the terrain and land softly, but also do it with a very high degree of precision.

4.3 CONCLUSIONS

Addressing the problem of integrated terrain-aware guidance for autonomous spacecraft landing, an algorithm to detect hazardous terrains and steer the spacecraft is presented. The proposed algorithm, OTBU, ingests terrain information from a single downward pointing camera in the form of a grid map registered to the world-coordinates. Using this information, along with current state information, OTBU generates terrain barriers



(a) Net fuel consumption

(b) Terminal landing velocity

(c) Terminal landing dispersion (red circle: 95% confidence)

Figure 4.4: Qualitative results of net fuel consumption, terminal velocity and terminal landing dispersion.

as polynomials at every pre-decided time-interval (based on the distance to the desired landing site). These terrain barriers are then used by MSS-OTALG to drive the spacecraft towards the desired landing site while avoiding the hazardous terrain. The effectiveness of online MSS-OTALG is demonstrated using extensive precision soft landing simulations in real Martian terrain taken from the HiRISE database. The results show that online MSS-OTALG is not only able to avoid the terrain but is also able to reach there with high accuracy as shown by Monte Carlo-like simulations.

CHAPTER 5

CONCLUSIONS

The motivation for this thesis was to develop guidance laws for planetary landing. Specifically, the goal was to develop guidance laws which can drive the spacecraft to land softly at the desired landing site with a high degree of precision and autonomously manoeuvre away from the terrain, all while consuming the least amount of fuel. To achieve these objective, first, a near-fuel-optimal guidance law for landing in hazardous terrain with minimal fuel consumption has been presented, titled "Optimal Terrain Avoidance Landing Guidance" (OTALG). The terrain has been modelled as n-step-shaped polygons, and barriers have been defined over the terrain model using piece-wise smooth polynomials. The efficacy of the guidance law is tested in a trench in the Martian environment, emulated using a flat-top, 2-step shape. The developed guidance algorithm successfully achieves precision soft landing, while also avoiding any collision with the terrain. Using extensive simulations, the near-fuel optimality is confirmed. The guidance law avoids terrain under bounded perturbation and maintains near-fuel optimality.

OTALG, while successful in soft landing while avoiding terrain, lacked in terms of robustness. The performance in terms of soft landing was severely degraded when uncertainties and disturbances were introduced. The OTALG law was subsequently expanded with the help of sliding mode control using multiple sliding surfaces (MSS) to improve the precision soft landing performance, and a near-fuel optimal and robust guidance law for precision soft-landing in hazardous terrain, named MSS-OTALG, was presented. The incorporation of MSS renders it the desired robustness. To allow the lander to manoeuvre away from the terrain, a state and time-dependent sliding parameter is introduced, and practical fixed time stability is proven under the proposed guidance law. Finally, extensive computer simulations validate the ability of the MSS-OTALG to avoid the terrain and precisely and softly land at the desired landing site while having low fuel consumption, under realistic limitations posed by thruster dynamics, thrust constraints and atmospheric disturbances. When compared against the OTALG and an augmented version of optimal sliding guidance (OSG) using Monte Carlo simulations, it was observed that the MSS-OTALG finds the middle ground in terms of all the performance measures.

In the discussions above, it is assumed that the terrain features are sufficiently and reliably available to the mission designers. However, this is not the case. The orbital images (which are generally the primary source for the terrain information of celestial bodies) are often plagued with imaging artefacts, shadows and nonuniform lighting. Thus to improve the autonomy of the landing guidance, a method to generate the terrain barriers by using the onboard sensed data, instead of relying solely on the pre-determined terrain information is proposed. To that end, an algorithm to detect hazardous terrains and steer the spacecraft is presented. The proposed algorithm, titled "Online Terrain Barrier Updates" (OTBU), uses the terrain information from a downward-pointing onboard camera and the current state information to generate terrain barriers as polynomials every pre-decided intervals (based on the distance to the desired landing site). Here, it is assumed that the pre-processed terrain information is available as a grid map, registered to the world-coordinates. The terrain barriers are then fed into MSS-OTALG law to drive the spacecraft towards the desired landing site for a terrain-avoided precision soft landing. The effectiveness of online MSS-OTALG is demonstrated using extensive precision soft landing simulations in real Martian terrain taken from the HiRISE database. The results show that online MSS-OTALG is not only able to avoid the terrain but is also able to reach there with high accuracy as shown by Monte Carlo-like simulations and statistical results.

5.1 SOME DIRECTIONS FOR FUTURE RESEARCH

The integrated terrain barrier generation and guidance law for planetary landing improves the state-of-the-art by being practically implementable and having low computational burden. However, there are certain directions which still require more investigation, as listed below:

- **Incorporation of spacecraft dynamics.** The current formulation is based on a 3DoF kinematic model of spacecraft, and a robust controller for stabilising roll and pitch rates is necessary to avoid excessive tumbling.

- **Development of the computer vision pipeline.** The OTBU algorithm currently assumes that the elevation map is readily available for it to ingest and generate the terrain barriers. However, for practical purposes it is necessary that a fast and accurate pipeline be developed, that can run onboard, and integrate it with the algorithm. Further, the pipeline may include the feature to detect a suitable landing site, or update a predetermined landing site if there is some obstacle, and update the guidance law accordingly.

- **Time-to-go estimation.** This thesis has relied on fixed terminal time to generate the guidance commands. However, selecting the fixed terminal time is heuristic process. Either a method may be developed to determine this final time which can incorporate the time taken by divert manoeuvres and update the time-to-go online, or the optimal control problem could be reformulated in a free final time setting.

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